

PROBLEME

1.5.1. $t_{ucpA} = 10s$ $f_A = 800 \text{ MHz}$
 $t_{ucpB} = 6s$ $C_{ucpB} = 1,2 \times C_{ucpA}$
 $f_B = ?$

$t_{ucp} = C_{ucp} \times t_c$
 $t_{ucp} = C_{ucp} \times \frac{1}{f}$

$t_{ucp} = C_{ucp} \times t_c = \frac{C_{ucp}}{f} \Rightarrow$

$\Rightarrow t_{ucpA} = \frac{C_{ucpA}}{f_A}$
 $10 = \frac{C_{ucpA}}{800}$
 $C_{ucpA} = 8000$

$t_{ucpB} = \frac{C_{ucpB}}{f_B}$
 $6 = \frac{1,2 \cdot C_{ucpA}}{f_B}$
 $f_B = \frac{1,2 \cdot 8000}{6} = 2 \cdot 800 = 1600 \text{ (MHz)}$

1.5.2. $t_{cA} = 4ns$ $CPI_A = 2$
 $t_{cB} = 2ns$ $CPI_B = 1,2$

Care e mai rapid?

$t_{ucp} = C_{ucp} \times t_c = \underbrace{N \times CPI \times t_c}$

$t_{ucpA} = N \times 2 \times 4 = 8N$
 $t_{ucpB} = N \times 1,2 \times 2 = 2,4N$ $\Rightarrow t_{ucpA} > t_{ucpB} \Rightarrow B \text{ mai rapid}$

1.5.3.

Clasă Instr.	CPI
A	1
B	2
C	3

Secv. cod	Contor instr. pt. clasa		
	A	B	C
S1	2	1	2
S2	4	1	1

$\Rightarrow N = 5$
 $\Rightarrow N = 6$

Care secv. e mai rapidă? CPI pt. frec. secv?

$t_{ucp} = t_c \times \sum_{i=1}^n (CPI_i \times I_i)$

$I_i = \text{nr. de exec. a instr. } i \text{ într-un pt.}$
 $CPI_i = \text{CPI pt. instr. } i$

S1: $t_{ucp} = t_c \times \sum_{i=1}^3 (CPI_i \times I_i) = t_c \times (CPI_A \times I_A + CPI_B \times I_B + CPI_C \times I_C)$
 $= t_c \times (1 \times 2 + 2 \times 1 + 3 \times 2)$
 $= t_c \times 10$

S2: $t_{ucp} = t_c \times (1 \times 4 + 2 \times 1 + 3 \times 1) = t_c \times 9$ $\left. \vphantom{t_{ucp}} \right\} S2 \text{ mai rap.}$

$CPI = \sum_{i=1}^n (CPI_i \times \frac{I_i}{N})$

S1 $CPI = 1 \times \frac{2}{5} + 2 \times \frac{1}{5} + 3 \times \frac{2}{5} = \frac{10}{5} = 2$

S2 $CPI = \frac{9}{6} = 1,5$

1.5.4.

Clasa	CPI
A	1
B	2
C	3

Cod generat de	Contoare instr. pt. clasa		
	A	B	C
Compilat. 1	$5 \cdot 10^6$	$1 \cdot 10^6$	$1 \cdot 10^6$
Compilat. 2	$10 \cdot 10^6$	$1 \cdot 10^6$	$1 \cdot 10^6$

$\rightarrow N = 7$
 $\rightarrow N = 12$

$$f = 600 \text{ MHz}$$

Ce secu. e mai rapidă conform MIPS?
 Dar conform t_E ?

$$\text{MIPS} = \frac{N}{t_E \cdot 10^6} = \frac{f}{\text{CPI} \cdot 10^6}$$

$$\text{CPI} = \sum_{i=1}^n (\text{CPI}_i \times \frac{\pm_i}{N})$$

compilator 1

$$\text{CPI} = 1 \times \frac{5 \cdot 10^6}{7 \cdot 10^6} + 2 \times \frac{1 \cdot 10^6}{7 \cdot 10^6} + 3 \cdot \frac{1 \cdot 10^6}{7 \cdot 10^6} = \frac{10}{7}$$

$$\text{MIPS} = \frac{600}{\frac{10}{7} \cdot 10^6} = \frac{42}{10^5} = 42 \cdot 10^{-5} = 420 \cdot 10^{-6}$$

compilator 2

$$\text{CPI} = 1 \times \frac{10 \cdot 10^6}{12 \cdot 10^6} + 2 \times \frac{1 \cdot 10^6}{12 \cdot 10^6} + 3 \times \frac{1 \cdot 10^6}{12 \cdot 10^6} = \frac{15}{12}$$

$$\text{MIPS} = \frac{f}{\text{CPI} \cdot 10^6} = \frac{400}{\frac{15}{12} \cdot 10^6} = \frac{4}{\frac{5}{4} \cdot 10^4} = \frac{16}{5} \cdot 10^{-4} \sim 3 \cdot 10^{-4}$$

$$420 \cdot 10^{-6} > 3 \cdot 10^{-4} \Rightarrow \text{compil. 1 e mai rapid}$$

compil. 1

$$\text{MIPS} = \frac{N}{t_E \cdot 10^6} \Rightarrow t_E = \frac{N}{\text{MIPS} \cdot 10^6} = \frac{7 \cdot 10^6}{420 \cdot 10^{-4} \cdot 10^6} = \frac{1}{6} \cdot 10^9$$

compil 2

$$t_E = \frac{12 \cdot 10^6}{\frac{16}{5} \cdot 10^{-4} \cdot 10^6} = \frac{15}{4} \cdot 10^9 = \frac{150}{4} \cdot 10^8 \sim 32,5 \cdot 10^8$$

$$\frac{1}{6} \cdot 10^9 < 32,5 \cdot 10^8 \Rightarrow \text{compil 2 e mai rapid}$$

① Comp. with 3 instruct. clases A, B, C with cpi values 1, 2, 3
 For a certain program compiler 1 generates a sequence of 4, 3, 2 million instructions. For the same program compiler 2 generates 6, 4, 2 million instr.
 Clock freq. is 2,4 GHz. What are the exec. times generated by the 2 compilers?

Class	CPI	Comp. 1	Comp 2	$f = 2,4 \text{ GHz}$
A	1	$4 \cdot 10^6$	$6 \cdot 10^6$	
B	2	$3 \cdot 10^6$	$4 \cdot 10^6$	
C	3	$2 \cdot 10^6$	$2 \cdot 10^6$	
		$\underbrace{\hspace{1cm}}_N$	$\underbrace{\hspace{1cm}}_N$	
		(contor instructiuni)		

Clasă	CPI
A	1
B	2
C	3

Cod. gen.	contoare instr. pt. $\leftarrow I_i$			
	A	B	C	
comp1	$4 \cdot 10^6$	$3 \cdot 10^6$	$2 \cdot 10^6$	$\rightarrow N = 9 \cdot 10^6$
comp2	$6 \cdot 10^6$	$4 \cdot 10^6$	$2 \cdot 10^6$	$\rightarrow N = 12 \cdot 10^6$

comp1

$$CPI = 1 \times \frac{4 \cdot 10^6}{9 \cdot 10^6} + 2 \cdot \frac{3 \cdot 10^6}{9 \cdot 10^6} + 3 \cdot \frac{2 \cdot 10^6}{9 \cdot 10^6} = \frac{16}{9}$$

$$MIPS = \frac{2400}{\frac{16}{9} \cdot 10^6} = \frac{21}{2} \cdot 10^{-4}$$

$$MIPS = \frac{N}{t_E \cdot 10^6} = \frac{f}{CPI \cdot 1}$$

$$CPI = \sum_{i=1}^n (CPI_i \cdot \frac{I_i}{N})$$

$$MIPS = \frac{N}{t_E \cdot 10^6} = \frac{9 \cdot 10^6}{t_E \cdot 10^6} \Rightarrow \left(t_E = \frac{9}{MIPS} = \frac{9}{\frac{21}{2} \cdot 10^{-4}} = \frac{6}{7} \cdot 10^4 \right)$$

comp2

$$CPI = 1 \cdot \frac{6 \cdot 10^6}{12 \cdot 10^6} + 2 \cdot \frac{4 \cdot 10^6}{12 \cdot 10^6} + 3 \cdot \frac{2 \cdot 10^6}{12 \cdot 10^6} = \frac{20}{12} = \frac{5}{3}$$

$$MIPS = \frac{f}{CPI \cdot 10^6} = \frac{2400}{\frac{5}{3} \cdot 10^6} = \frac{72}{5} \cdot 10^{-4}$$

$$MIPS = \frac{N}{t_E \cdot 10^6} \Rightarrow \left(t_E = \frac{N}{MIPS \cdot 10^6} = \frac{12 \cdot 10^6}{\frac{72}{5} \cdot 10^{-4} \cdot 10^6} = \frac{5}{6} \cdot 10^4 \right)$$

2

Clasa	CPI
A	1
B	2
C	3

Cod. generat.	I_i			
	A	B	C	
comp 1	$5 \cdot 10^6$	$2 \cdot 10^6$	$3 \cdot 10^6$	$\rightarrow 10.11$
comp 2	$10 \cdot 10^6$	$4 \cdot 10^6$	$2 \cdot 10^6$	$\rightarrow 16.11$

$t_E = ?$ MIPS = ?

! Frecu. ceasului nu e dată.
Vom folosi variabl. f .

$$\text{MIPS} = \frac{f}{\text{CPI} \cdot 10^6}$$

$$\text{CPI} = \sum_{i=1}^n (\text{CPI}_i \times \frac{I_i}{N})$$

compil. 1

$$\text{CPI} = 1 \times \frac{5 \cdot 10^6}{10 \cdot 10^6} + 2 \times \frac{2 \cdot 10^6}{10 \cdot 10^6} + 3 \times \frac{3 \cdot 10^6}{10 \cdot 10^6} = \frac{18}{10} = 1.8$$

GREȘEALĂ
 $5 + 4 + 9 = 18$
 $\text{CPI} = \frac{18}{10} =$

$$\text{MIPS} = \frac{f}{\text{CPI} \cdot 10^6} = \frac{f}{1.8 \cdot 10^6} = \frac{10}{18} \cdot 10^6 \cdot f = \frac{5}{9} \cdot 10^{-6} \cdot f$$

$$\text{MIPS} = \frac{N}{t_E \cdot 10^6} \Rightarrow t_E = \frac{N}{\text{MIPS} \cdot 10^6} = \frac{10^2 \cdot 10^6}{\frac{5}{9} \cdot 10^6 \cdot 10^6 \cdot f} = \frac{18 \cdot 10^6}{f} \cdot \frac{1}{f}$$

compil 2

$$\text{CPI} = 1 \times \frac{10 \cdot 10^6}{16 \cdot 10^6} + 2 \times \frac{4 \cdot 10^6}{16 \cdot 10^6} + 3 \times \frac{2 \cdot 10^6}{16 \cdot 10^6} = \frac{24}{16} = \frac{3}{2} = 1.5$$

$$\text{MIPS} = \frac{f}{\text{CPI} \cdot 10^6} = \frac{f}{1.5 \cdot 10^6} = \frac{2}{3} \cdot f \cdot 10^{-6}$$

$$\text{MIPS} = \frac{N}{t_E \cdot 10^6} \Rightarrow t_E = \frac{N}{\text{MIPS} \cdot 10^6} = \frac{16 \cdot 10^6}{\frac{2}{3} \cdot f \cdot 10^6 \cdot 10^6} = \frac{24 \cdot 10^6}{f}$$

1.5.8.

PROBLEME

Program	Op. in VM	Timep. exec. (secunde)		
		A	B	C
P ₁	10.000.000	1	10	20
P ₂	100.000.000	1000	100	20

A, B, C calc. dig
care e mai r
conforme t_E
total?

$$t_{E_{total}}(A) = 1001 \quad t_{E_{total}}(B) = 110 \quad t_{E_{total}}(C) = 40 \Rightarrow C \text{ cel. ma}$$

15.9. Det. MFLOPS pt. P₁ & P₂. P_p ca' fice, op. ore ponde

P₁

$$\begin{aligned} \text{MFLOPS}(A) &= \frac{10 \cdot 10^6}{1 \times 10^6} = 10 \\ \text{MFLOPS}(B) &= \frac{10 \cdot 10^6}{10 \times 10^6} = 1 \\ \text{MFLOPS}(C) &= \frac{10 \cdot 10^6}{20 \times 10^6} = 0,5 \end{aligned}$$

$$\text{MFLOPS} = \frac{N_{VM}}{t_E \cdot 10}$$

$$N_{VM} = \text{nr. op. VM din}$$

P₂

$$\begin{aligned} \text{MFLOPS}(A) &= \frac{100 \cdot 10^6}{1000 \times 10^6} = 0,1 \\ \text{MFLOPS}(B) &= \frac{100 \cdot 10^6}{100 \times 10^6} = 1 \\ \text{MFLOPS}(C) &= \frac{100 \cdot 10^6}{20 \times 10^6} = 5 \end{aligned}$$

15.10. Comparatie utiliz. M.G. a t_E .

$$MG = \sqrt[n]{\prod_{i=1}^n t_{Ei}} \quad t_{Ei} = \text{timep exec. normalizat}$$

Program	$t_E(A)$	$t_E(B)$	$t_E(C)$	Norm. A			Norm B			Norm C	
				$t_E(A)$	$t_E(B)$	$t_E(C)$	A	B	C	A	B
P ₁	1	10	20	1	$\frac{10}{1} = 10$	$\frac{20}{1} = 20$		1			
P ₂	1000	100	20	1	$\frac{100}{1000} = 0,1$	$\frac{20}{1000} = 0,02$		1			
M.G.	31,6	31,6	6,3	1	1	$\frac{6,3}{31,6} \approx 0,2$		1			

$$MG(A) = \sqrt[3]{1000 \cdot 1} \approx 31,6$$

$$MG(B) = \sqrt[3]{100 \cdot 10} \approx 31,6$$

5.5. Calc. → cu unit. de calcul în VM (clasele din tabel)
 → fără unit. (emulează instr. în VM)

Instr.	Frecv.
Adun. VM	15%
Inmult. VM	10%
Împart. VM	5%
Instr. întregi	70%

Instr.	Nr. cicluri de ceas	
	Cu unit. VM	Fără unit. VM
Adun VM	4	40
Inm. VM	6	60
Împ. VM	20	100
Instr. întregi	2	2

$$f = 600 \text{ MHz}$$

MIPS pt. ambele?

$$CPI = \sum_{i=1}^n (CPI_i \times F_i)$$

cu unit. VM

$$CPI = 4 \cdot 0,15 + 6 \cdot 0,1 + 20 \cdot 0,05 + 2 \cdot 0,7 = 0,6 + 0,6 + 1 + 1,4 = 3,6$$

$$MIPS = \frac{f}{CPI \times 10^6} = \frac{600 \text{ MHz}}{3,6 \times 10^6} = 0,166 \cdot 10^{-4} \cdot 10^6 = 16,6$$

fără unit. VM

$$CPI = 40 \cdot 0,15 + 60 \cdot 0,1 + 100 \cdot 0,05 + 2 \cdot 0,7 = 6 + 6 + 5 + 1,4 = 18,4$$

$$MIPS = \frac{f}{CPI \times 10^6} = \frac{600 \text{ MHz}}{18,4 \times 10^6} \approx 0,32 \cdot 10^{-4} \cdot 10^6 \approx 32$$

15.6. Pt. un progr. P pe calc. cu unit. VM → $\overbrace{300 \cdot 10^6}^N$ inst
 Câte instr. întregi necesită calc. fără unit. în VM

cu VM

$$\text{Instr. întregi} \quad \frac{70}{100} = \frac{I_i}{300 \cdot 10^6} \quad F_i = \frac{I_i}{N}$$

$$\Rightarrow I_i = 210 \cdot 10^6$$

fără VM

$$\frac{70}{100} = \frac{210 \cdot 10^6}{N} \Rightarrow \frac{210 \cdot 100}{70} \cdot 10^6 = 300 \cdot 10^6$$

! Același N pt. că se poate observa în tabel că instr. necesită același nr. de cicluri de ceas.

cu VM

$$N = 300 \cdot 10^6$$

$$\text{Instr. Intregi } 70\% \cdot N \Rightarrow \dot{I}_{\text{Intregi}} = 70\% \cdot 300 \cdot 10^6 = 210 \cdot 10^6$$

fără VM

$$\text{Instr. Intregi } 40\% \cdot N$$

VM:

$$I_i = \frac{\dot{I}_i}{N}$$

Instr.	Frecv.	$\dot{I}_i$ cu VM
+	15%	$45 \cdot 10^6$
*	10%	$30 \cdot 10^6$
/	5%	$15 \cdot 10^6$
Intregi	70%	$210 \cdot 10^6$
TOTAL	100%	$300 \cdot 10^6$

$$t_{ucp} = \frac{N * CPI}{f} = \frac{300 \cdot 10^6 \times 3,6}{600 \cdot 10^6} = 1,8$$

fără VM

$$CPI = \frac{\sum_{i=1}^n (CPI_i * I_i)}{N}$$

$$\Rightarrow N = \frac{40 \times 45 \cdot 10^6 + 60 \times 30 \cdot 10^6 + 20 \cdot 10^6}{32}$$

$$N = \frac{180 + 180 + 150 + 20}{32}$$

$$N = \frac{530}{32} \cdot 10^6$$

$$N \approx 16,5 \cdot 10^6$$

imposibil

1.5.11

Calc. + mod vectorial $\Rightarrow t_E$ 20 de ori mai redusProcentaj vectorizare = $\% \text{ din timp}$ se poate utiliza modul vect.
Ce proc. vectoriz e necesar pt. creșt vitezei de 2 ori?exec. în mod vect. $\Rightarrow t_E$ de 20 ori redus $\Rightarrow \Delta v_{inub} = 20$
creșterea vitezei de 2 ori $\Rightarrow \Delta v_{tot} = 2$ procentaj vectoriz = F_{inub} (fract. timpului de exec. când se poate beneficia de îmbunătăț)

$$t_{E_{nou}} = t_{E_{init}} * \left((1 - F_{inub}) + \frac{F_{inub}}{\Delta v_{inub}} \right)$$

$$\underbrace{\Delta v_{tot}}_2 = \frac{t_{E_{init}}}{t_{E_{nou}}} = \frac{1}{1 - F_{inub} + \frac{F_{inub}}{\Delta v_{inub}}} \quad (=) \quad 1 - F_{inub} + \frac{F_{inub}}{20} = \frac{1}{2}$$

$$\frac{20 - 20F_{inub} + F_{inub}}{20} = \frac{1}{2}$$

$$-19F_{inub} = 10 - 20$$

$$F_{inub} = \frac{10}{19} \sim 0,5$$

15.13

Tip instr.	Frecv.	CPI
A	40%	2
B	25%	3
C	25%	3
D	10%	5

$$MIPS = ? \quad f = 800 \text{ MHz}$$

$$MIPS = \frac{800 \cdot 10^6}{2,8 \cdot 10^6} = \frac{100}{4} \sim 14,2$$

$$MIPS = \frac{f}{CPI \cdot 10^6}$$

$$CPI = \sum_{i=1}^n (CPI_i \cdot F_i)$$

$$= 2 \times 0,4 + 3 \times 0,25 + 3 \times 0,25 + 5 \times 0,1$$

$$= 0,8 + 0,75 + 0,75 + 0,5$$

$$= 2,8$$

PROBLEMIE

1.5.14

Instr.	Frect.	CPI
* VM	10%	6
+ VM	15%	4
/ VM	5%	20
Instr. VF	70%	2

$$f = 800 \text{ MHz}$$

- frec. op. în VM are ponderea 1
- calc. exec. $300 \cdot 10^6$ instr. pt un program

$$\text{MFLOPS} = ?$$

$$\text{MFLOPS} = \frac{N_{VM}}{\text{MIPS} \cdot 10^6}$$

N_{VM} - număr instr. în VM

$$N_{VM} = (10\% + 15\% + 5\%)N = 30\% \cdot N$$

$$N_{VM} = 30\% \cdot 300 \cdot 10^6 = 90 \cdot 10^6$$

$$\text{MIPS} = \frac{f}{\text{CPI} \cdot 10^6}$$

$$\text{CPI} = \sum_{i=1}^n (\text{CPI}_i \cdot F_i)$$

$$= 6 \times 10\% + 4 \cdot 15\% + 20 \times 5\% + 2 \times 70\%$$

$$= 60\% + 60\% + 100\% + 140\%$$

$$= 3,6$$

$$\text{MIPS} = \frac{800 \cdot 10^6}{3,6 \cdot 10^6} = 250$$

$$\text{MFLOPS} = \frac{90 \cdot 10^6}{250 \cdot 10^6} = \frac{9}{25}$$

1.5.15 Îmbun. $\Rightarrow$ vit. exec. a instr. VM crește de 5 ori
 Timp. exec. a unui progr. de eval e 10 s inițial.
 Care e creșt vit de exec. ? 50% din timp exec.
 e consumat de instr. în VM.

$$\Delta V_{\text{imb}} = 5$$

$$F_{\text{imb}} = 50\% = 0,5$$

$$\Delta V_{\text{tot}} = \frac{1}{(1 - F_{\text{imb}}) + \frac{F_{\text{imb}}}{\Delta V_{\text{imb}}}} = \frac{1}{1 - 0,5 + \frac{0,5}{5}}$$

$$\Delta V_{\text{tot}} = \frac{1}{0,5 + 0,1} = \frac{10}{6} \sim 1,67$$

3.5.8. Deduceți expr. gen. pt. T_k pt. un SAT.

$$T_k = G_{0,k-1} + P_{0,k-1} \cdot T_0$$

$\uparrow$ toate ce cont., cel puțin un g $\uparrow$ produs propagare

$$G_{0,k-1} = g_{k-1} + p_{k-1} g_{k-2} + \dots +$$

$$P_{0,k-1} = p_{k-1} p_{k-2} \dots p_0$$

3.5.9. Deduceți T_{12} și T_{16} într-un SAT. Adunați 2 nr. de 16 biți. Nr. într. e 4.

$$T_4 = G_{0,3} + P_{0,3} \cdot T_0$$

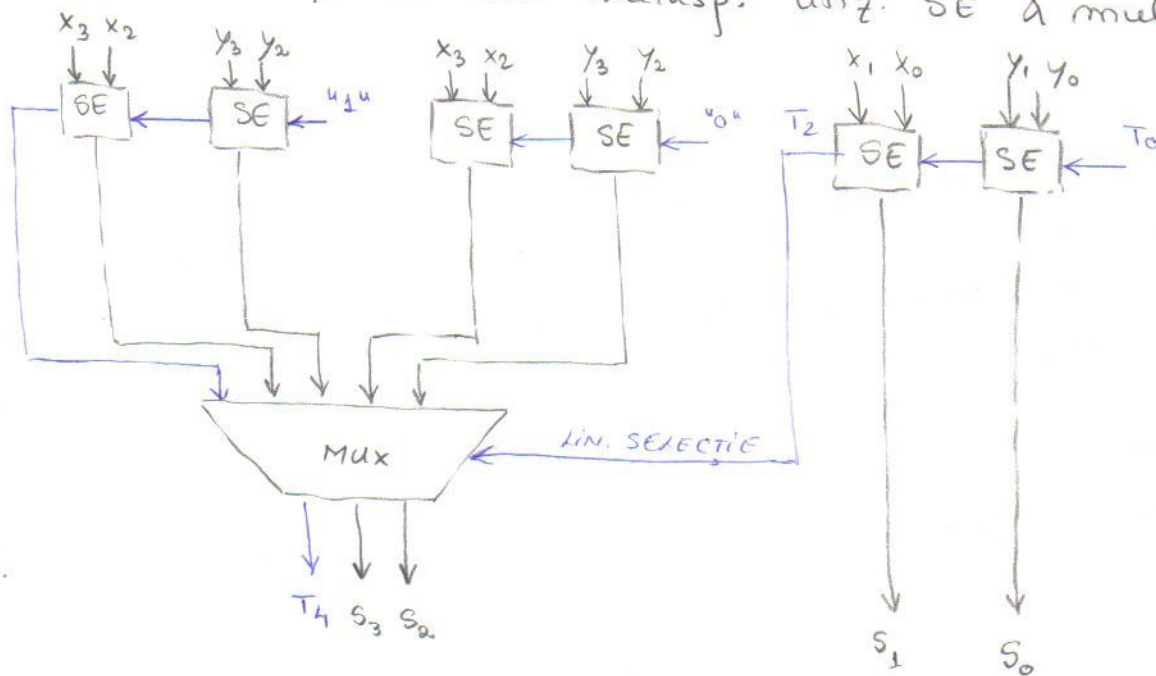
$$T_8 = G_{4,7} + P_{4,7} \cdot G_{0,3} + P_{4,7} \cdot P_{0,3} \cdot T_0$$

$$T_{12} = G_{8,11} + P_{8,11} \cdot G_{4,7} + P_{8,11} \cdot P_{4,7} \cdot G_{0,3} + P_{8,11} \cdot P_{4,7} \cdot P_{0,3} \cdot T_0$$

$$T_{16} = G_{12,15} + P_{12,15} \cdot G_{8,11} + P_{12,15} \cdot P_{8,11} \cdot G_{4,7} + P_{12,15} \cdot P_{8,11} \cdot P_{4,7} \cdot G_{0,3} + P_{12,15} \cdot P_{8,11} \cdot P_{4,7} \cdot P_{0,3} \cdot T_0$$

$\underbrace{\hspace{15em}}_{G_{0,15}} \quad \underbrace{\hspace{15em}}_{P_{0,15}}$

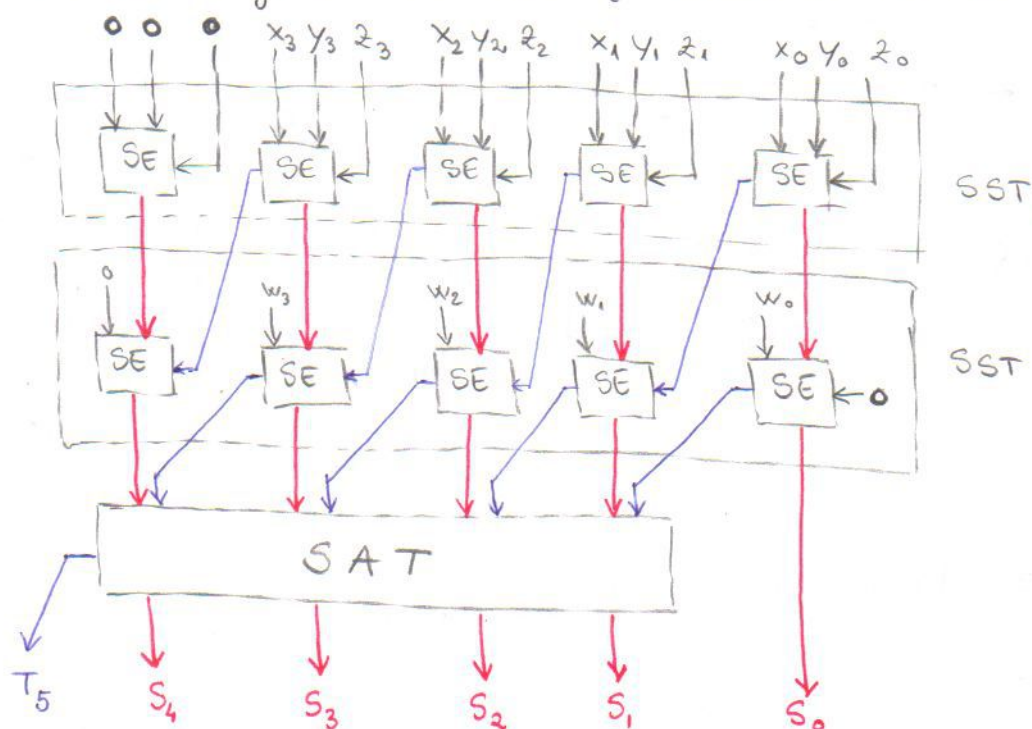
3.5.11. Sumat. 4 biți cu sel. transp. utiliz. SE & multiple.



3.5.12

Sumat. cu salu. transp. 4 nr. de 4 biți

Schema logică cu SE. ! Suma a 4 nr / 4 biți $\rightarrow n+2$ biți

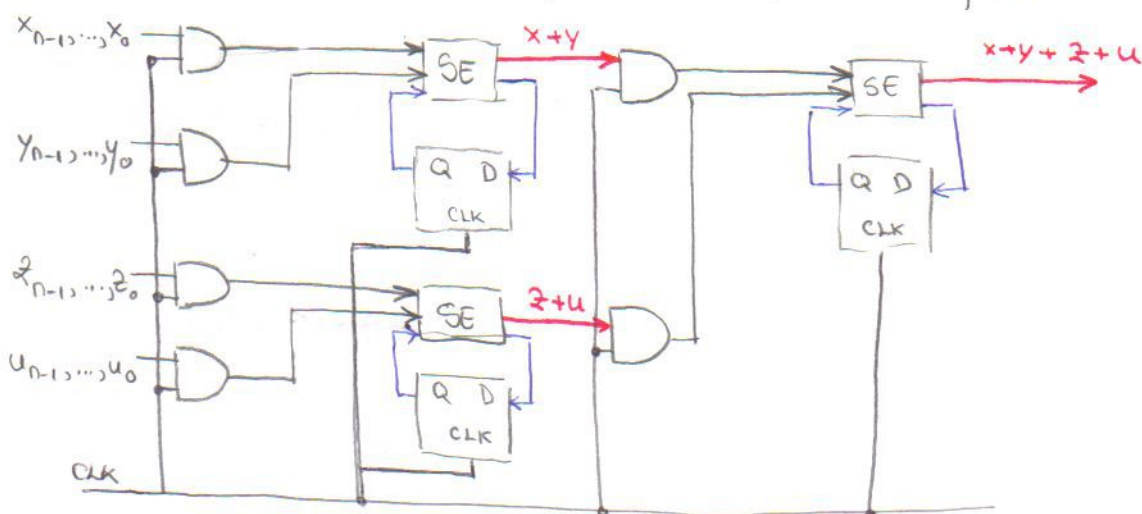


3.5.14

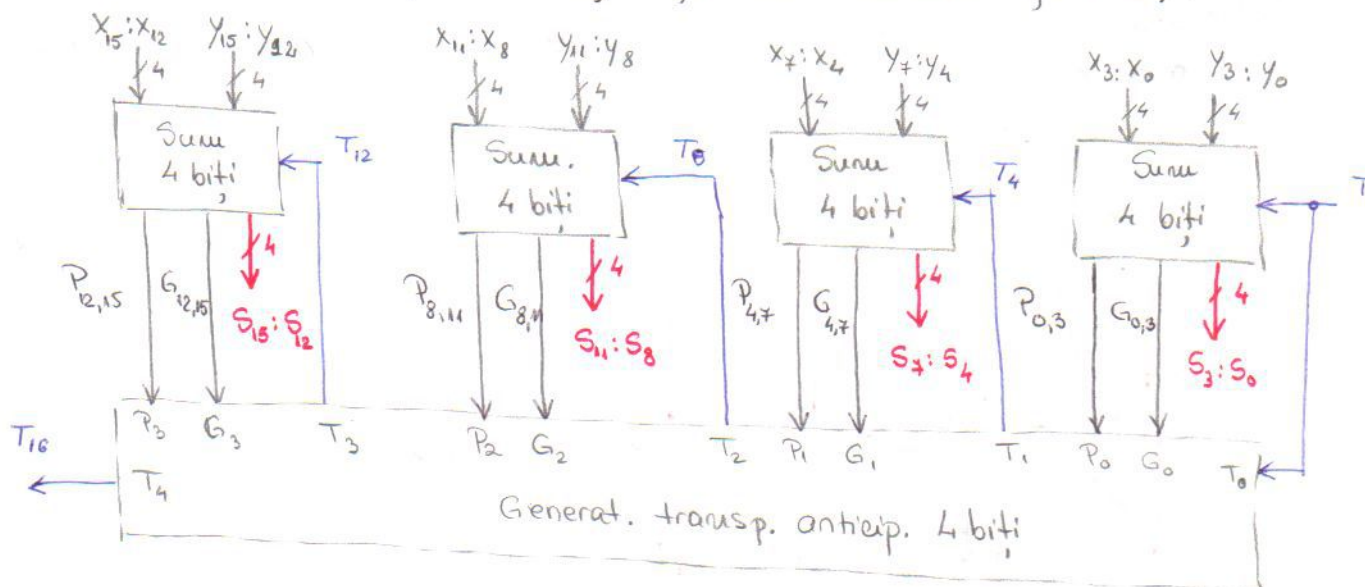
Sumat. serial : 4 nr. fără semn

Are 4 linii intrare x, y, z, u
1 linie ieseire s

Numerele încep cu bitii c.m.p.s



5) Proiect. un SAT pe 16 biti folosind sumat. pe 4 biti.



Det. ec. de transport pt. fiecare circuit.

$$T_4 = G_{0,3} + P_{0,3} \cdot T_0$$

$$G_{0,3} = g_3 + p_3 g_2 + p_3 p_2 g_1 + p_3 p_2 p_1 g_0$$

$$P_{0,3} = p_3 p_2 p_1 p_0$$

$$T_8 = G_{4,7} + P_{4,7} \cdot T_4$$

$$G_{4,7} = g_7 + p_7 g_6 + p_7 p_6 g_5 + p_7 p_6 p_5 g_4$$

$$P_{4,7} = p_7 p_6 p_5 p_4$$

$$T_{12} = G_{8,11} + P_{8,11} T_8$$

$$G_{8,11} = g_{11} + p_{11} g_{10} + p_{11} p_{10} g_9 + p_{11} p_{10} p_9 g_8$$

$$P_{8,11} = p_{11} p_{10} p_9 p_8$$

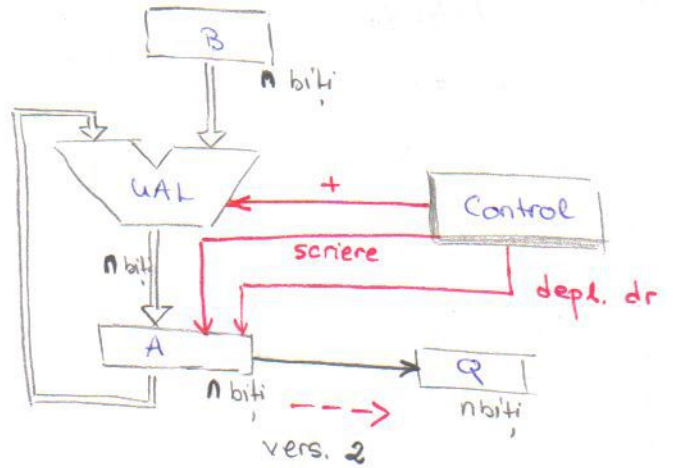
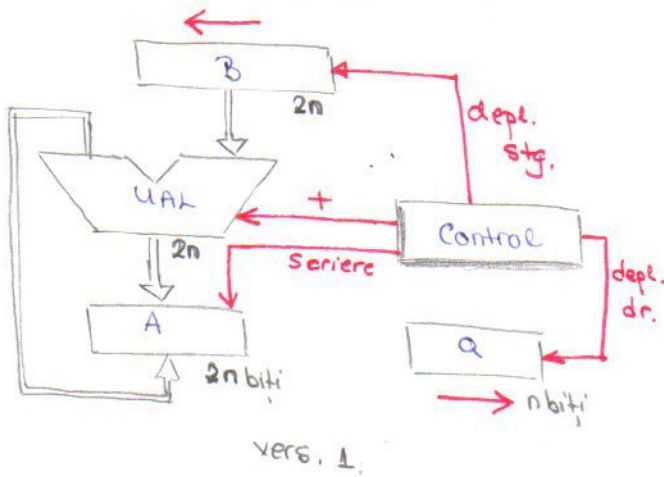
$$T_{16} = G_{12,15} + P_{12,15} T_{12}$$

$$G_{12,15} = g_{15} + p_{15} g_{14} + p_{15} p_{14} g_{13} + p_{15} p_{14} p_{13} g_{12}$$

$$P_{12,15} = p_{15} p_{14} p_{13} p_{12}$$

! In carte pag. 82 e schema unui sumat. cu antcip. de 4 biti

7NM. PRIN ADUN & DEPLAS

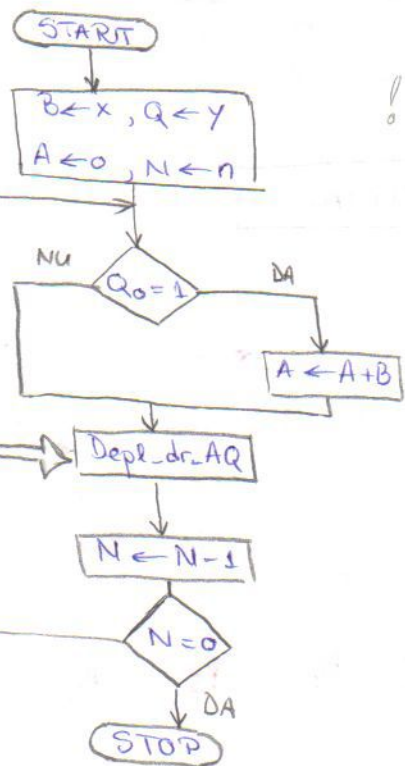
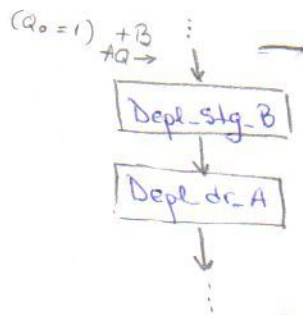


ex: vers 2

$$9 \times 12 = \overset{\text{depl. nm}}{1001} \times \overset{\text{norm.}}{1100}$$

PAS	A	Q	B
0	0000	1100	1001
1	0000	0110	
2	0000	0011	
3	0000	0011	
	1001 +		
	1001		
	0100	1001	
4	0100 +		
	1001		
	1101		
	0110	1100	

($Q_0=0$) $\Rightarrow$ depl. dr AQ
($Q_0=0$) $\Rightarrow$ AQ $\rightarrow$



MET. BOOTH

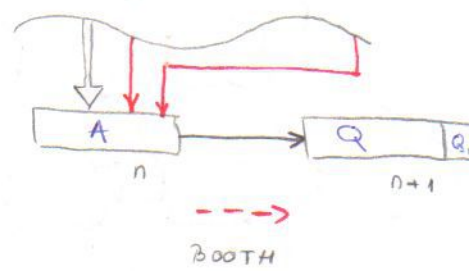
PAS	A	Q	Q-1	B
0	00000	10110	0	01101
1	00000	01011	0	
2	10011			
3	11001	10101	1	
4	11100	11010	1	
	01101			
	01001			
5	00100	11101	0	
	10011			
	10111			
	11011	11101		

$$x = 13 \quad y = -10$$

$$+13 = 01101 \text{ (B)} \quad -13 = 10011$$

$$-10 = -(1010) = 2(1010) = 10110$$

$-B \&$
 $\rightarrow$
 $\rightarrow$
 $\rightarrow$
 $+B \&$
 $\rightarrow$
 $-B \&$
 $\rightarrow$



00	$\rightarrow$
01	$+B \& \rightarrow$
10	$-B \& \rightarrow$
11	$\rightarrow$

$$110111110 = -(10000010) = -130$$

Rez: ~~110111110~~

7NM. BAZĂ SUP.

BOOTH în bază 4

$$Y_4 = \begin{matrix} & +1 \\ & \downarrow \\ Y_2 & Y_1 & Y_0 \\ & -3 \\ & \downarrow \\ & -1 \end{matrix}$$

nr. în bază 4

$$\begin{matrix} +1 \\ 1 & 0 & 3 & 2 & 0 \\ & \uparrow \\ & -1 \end{matrix} = 11(-1)20$$

$$\begin{matrix} 1 & 0 & 3 & 2 & 0 \\ 4 & 3 & 2 & 1 & 0 \end{matrix}$$

Y_{2i+1}	Y_{2i}	Y_{2i-1}
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

	2 poz
$\rightarrow$	$\rightarrow$
$+x$	$\rightarrow$
$+x$	$\rightarrow$
$+2x$	$\rightarrow$
$-2x$	$\rightarrow$
$-2x$	$\rightarrow$
$-x$	$\rightarrow$
$-x$	$\rightarrow$

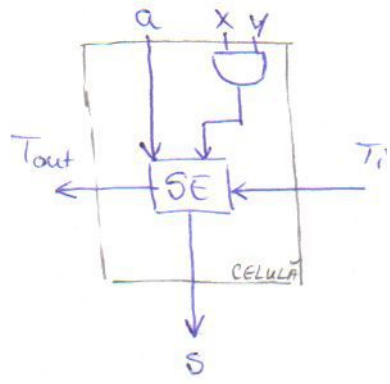
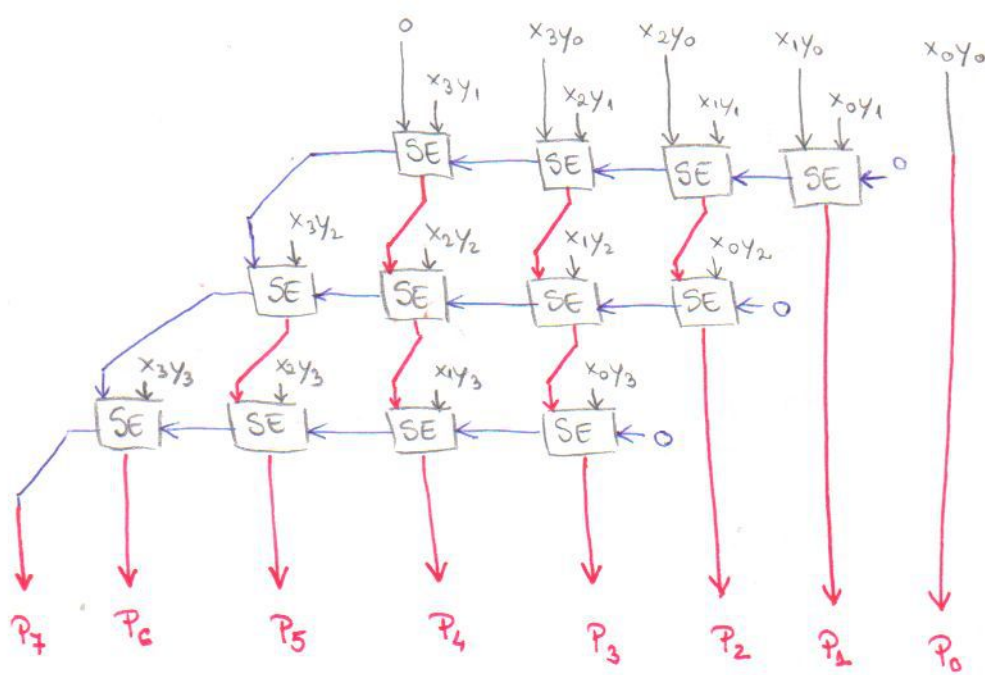
$$\begin{aligned} &= 1 \times 4^4 + 0 \times 4^3 + 3 \times 4^2 + 2 \times 4^1 + 0 \times 4^0 \\ &= \dots + 0 \times 4^3 + (4-1) \times 4^2 + \dots \\ &= \dots + 1 \times 4^3 + (-1) \times 4^2 + \dots \Leftrightarrow \end{aligned}$$

$$\begin{matrix} -1 \\ +4 \end{matrix}$$

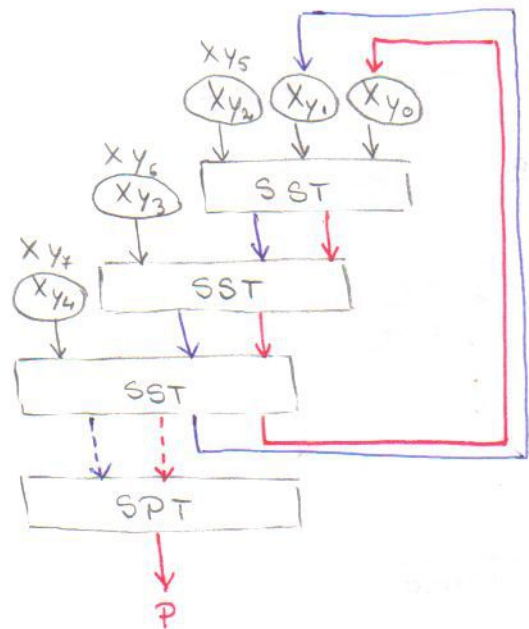
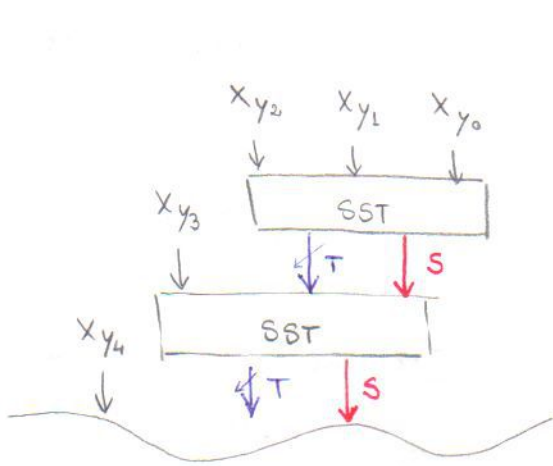
7NM. MATRICIALĂ

	x_3	x_2	x_1	x_0	*		
y_3	x_3y_0	x_2y_0	x_1y_0	x_0y_0			
y_2	x_3y_1	x_2y_1	x_1y_1	x_0y_1			
y_1	x_3y_2	x_2y_2	x_1y_2	x_0y_2			
y_0	x_3y_3	x_2y_3	x_1y_3	x_0y_3			
P_7	P_6	P_5	P_4	P_3	P_2	P_1	P_0

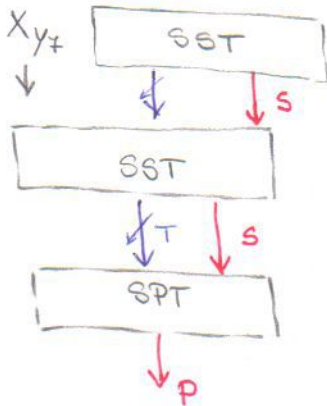
$$\begin{aligned} P_0 &= x_0y_0 \\ P_1 &= x_1y_0 + x_0y_1 \\ P_2 &= x_2y_0 + x_1y_1 + x_0y_2 \\ P_3 &= x_3y_0 + x_2y_1 + x_1y_2 + x_0y_3 \\ P_4 &= x_3y_1 + x_2y_2 + x_1y_3 \\ P_5 &= x_3y_2 + x_2y_3 \\ P_6 &= x_3y_3 \end{aligned}$$



$$T_{out} S = a \text{ plus } xy \text{ plus}$$

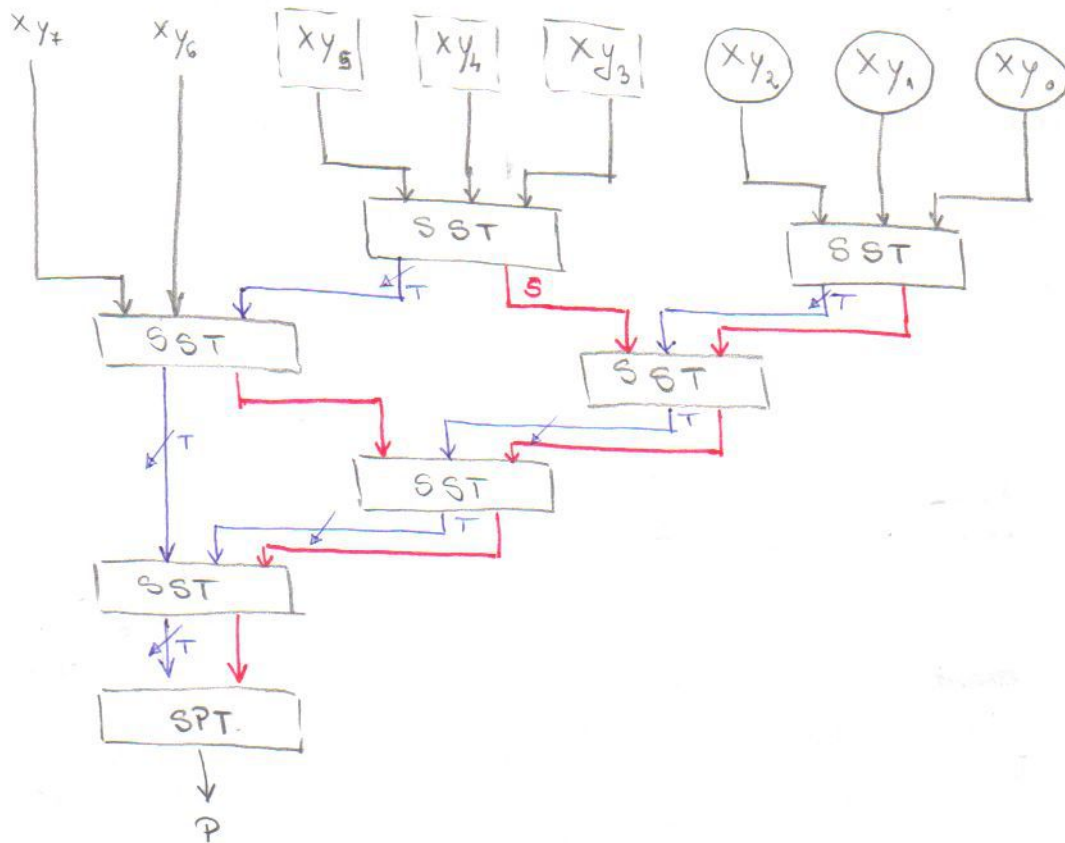


Cu 2 treceri!



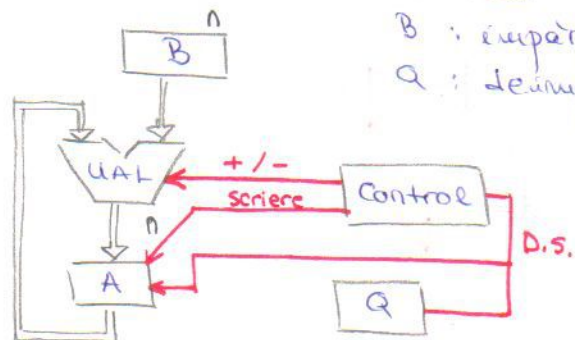
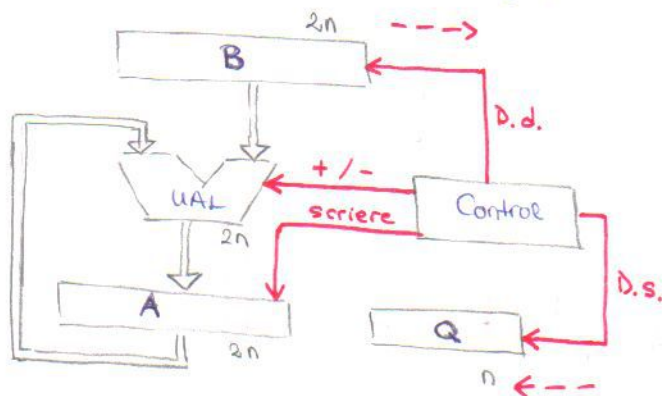
Inm. matric. cu SST

ARBORE WALLACE

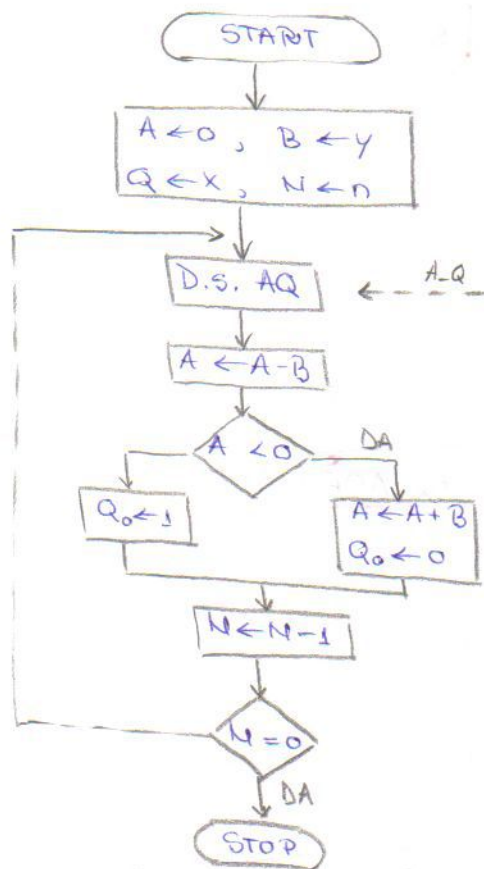
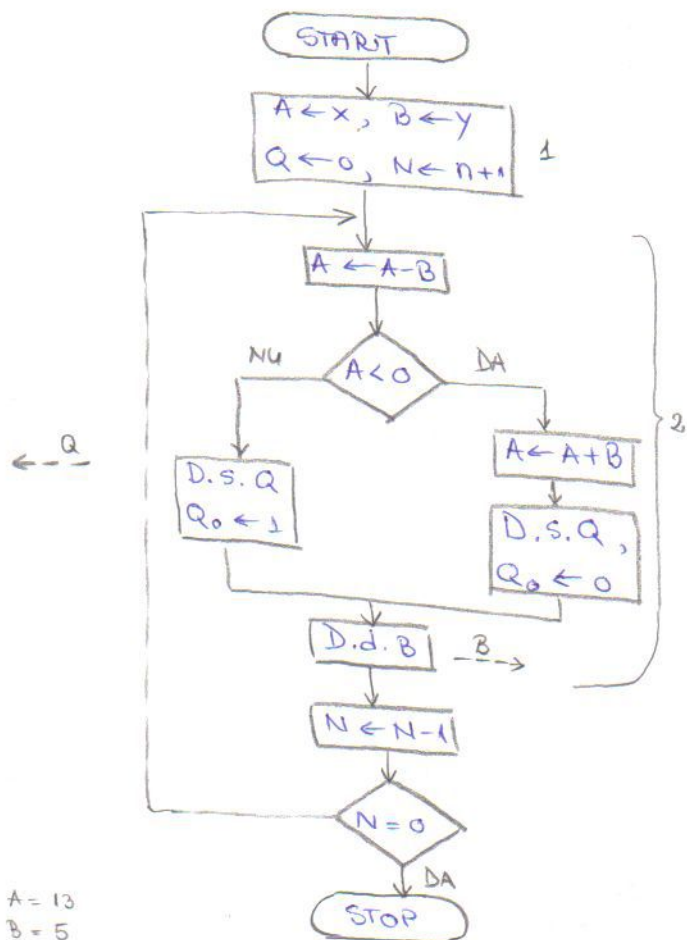


IMPĂRȚIREA

• CU REFAC. RESTULUI PARȚIAL



A: decomp & rest
B: împărțit.
Q: cât
A: rest
B: împărțit.
Q: decomp & cât



A = 13
B = 5

PAS	A	Q	B
0	00001101	0000	01010000
1	10111101	0000	A ← A + B
	00001101	0000	A ← A + B; Q; Q0 ← 0
2	11011000	0000	B ← 00101000
	11011000	0000	A ← A - B
	11011000	0000	A ← A + B; Q; Q0 ← 0
3	...	...	B ← 00010100
5	...	...	...
	rest 0011	cât 0010	00000010

PAS	A	Q	B
0	00000	1101	0101
1	00001	1010	
2	00001	1010	
3	00011	1000	
4	00011	1000	
5	00011	1000	
6	00001	1001	

-B = 11011
BCC2)

AQ
A ← A - B
A ← A + B, Q0 ← 0
AQ
A ← A - B
A ← A + B, Q0 ← 0
AQ
A ← A - B
Q0 ← 1

• FĂRĂ REFACERE

1. $R \leftarrow R - y$

$Rez < 0 \Rightarrow R \leftarrow R - y + y$ (refacere rest)

2. $R \leftarrow 2 \times R$ (depl. stg. R cu o poziție)

$R \leftarrow 2 \times R - y$

sau $R \leftarrow R - y$

$Rez < 0 \Rightarrow R \leftarrow 2 \times R - 2 \times y$ (depl. stânga)

$R \leftarrow 2 \times R - 2 \times y + y$

! Depl. stânga cu o
poz. echiv. cu înmulț
cu 2

• MET. SRT

- bazată pe depl. peste zerouri

- algoritmul: ① Reg B conț. k zerouri pe pozi c.m.s
 $\Rightarrow$ depl. stânga k poziții (toate neg.)

② urmărit. de n ori

2.1 dacă 3 biti c.m.s din A sunt identici
 $\Rightarrow \checkmark A-Q ; Q_0 \leftarrow 0$

2.2 dacă 3 biti c.m.s din A nu sunt identici
& A negativ

$\Rightarrow \checkmark A-Q ; Q_0 \leftarrow 1 ; A \leftarrow A+B$

2.3 contrar

$\Rightarrow \checkmark A-Q ; Q_0 \leftarrow 1 ; A \leftarrow A-B$

ex: $x = 8$ (1000)
 $y = 3$ (0011)

PAS	A	Q	B	N
0	00000	1000	0011	4
1	00010	0000	1100	4
2	00100	0000	1100	3
3	01000	0000		2
	10100 +			
	11100	0001		
4	11000	0010		1
5	10000	0100		0
	01100	0101		
	11100			
6	01100 +	1+		
	01000	0010		
7	00010	0010	1100	

- (1100) =
1(0100)

$\checkmark A-Q ; \checkmark B$ (2 poz.)

$\checkmark A-Q ; Q_0 \leftarrow 0$

$\checkmark A-Q ; Q_0 \leftarrow 1 ; A \leftarrow A-B$

$\checkmark A-Q ; Q_0 \leftarrow 0$

$\checkmark A-Q ; Q_0 \leftarrow -1 ; A \leftarrow A+B$ ($Q_0 \leftarrow 1$)

$A < 0 \Rightarrow$ corectie $A \leftarrow A+B ; Q_0 \leftarrow Q_0 - 1$

$A \searrow$ (2 poz.)

③ dacă rest final < 0

$\Rightarrow$ corectare rest $A \leftarrow A + B$
cat $Q \leftarrow Q - 1$

④ * depl. rest dr $\rightarrow$ cu k poziții (cele inițiale)

19 Schema logică detaliată a unui circ. de înmulț pe 4 biți
cu organiz. arbore Wallace, utiliz. SE ca compon. ba

$$x y_0 \text{ pt. 4 biți} = x_0 y_0 + x_1 y_0 + x_2 y_0 + x_3 y_0$$

PROBLEME

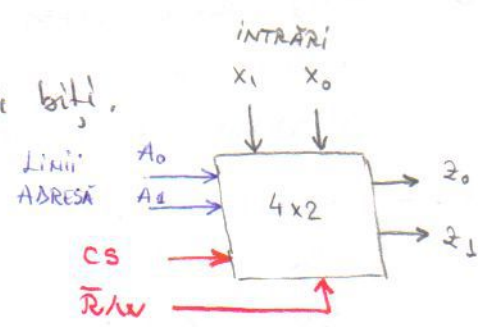
4.9.1. utiliz. unit. de mem. 4×2 biti.
Proiect. o mem. cu acces aleator 16×4 biti.

$$N = 4 \quad w = 2$$

$$N' = 16 \quad w' = 4$$

$$p = \frac{N'}{N} = \frac{16}{4} = 4$$

$$q = \frac{w'}{w} = \frac{4}{2} = 2$$



$$2^m \times w = 2^2 \times 2$$

- necesare $p \times q = 4 \times 2$ circuite de 4×2 biti
- $N' > N \Rightarrow$ necesare circuite pt decodif. adresei

$p = 4 \Rightarrow$ nr. adrese creste de 4 ori $\Rightarrow$ 2 linii suplimentare, ($4 = 2^2$)

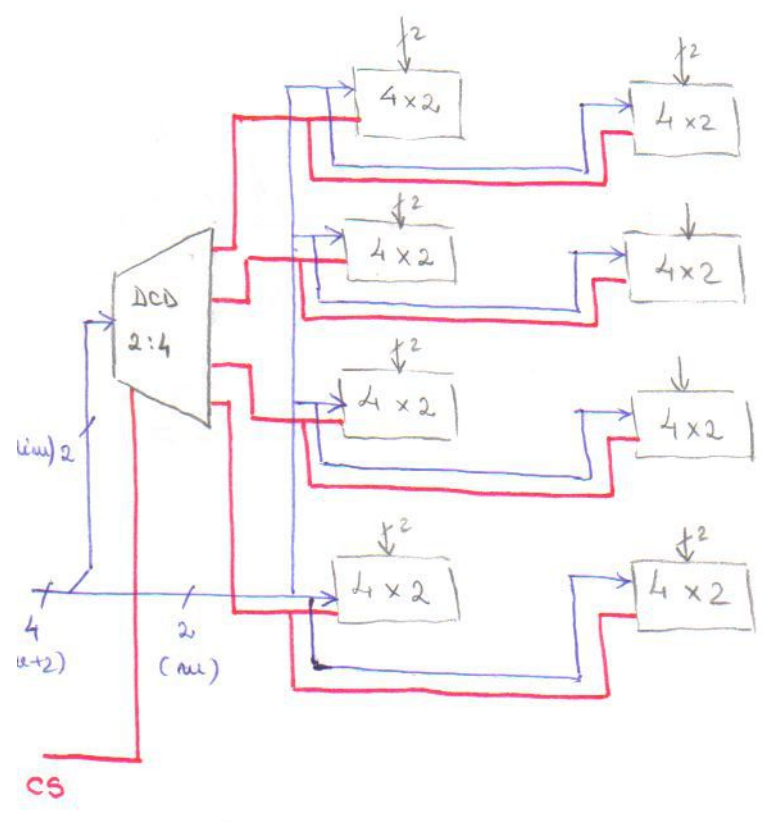
$q = 2 \Rightarrow$ cuvântul creste de 2 ori

$$2^m \times w = 2^2 \times 2$$

$$m = 2$$

$$m + 2 = 4$$

↑
suplim.



4.9.3 utiliz. mem. DRAM de 64 Mbiti (8E1) din sect. 4.4.4
 Proiect. o mem. DRAM 256 M x 32 biti. Desenati
 in stilul fig. 4.9 si 4.10. (ca si ex. 4.9.1 de mai devreme)

- 8E1: - capacitate 64 Mbiti (2^{26} biti, date)
 - info 2^{23} octeti
 - 8M x 8 biti ($m = 23$ $w = 8$)
 - adresa: • linie 13 biti
 • coloana 10 biti

$$256 = 4 \times 64 \text{ M} \quad \underbrace{\quad}_{2^2}$$

- 13 linii exterioare pt. adresă $\Rightarrow$ adr lin & col mult
 - $\overline{\text{RAS}}$ & $\overline{\text{CAS}}$

$$2^m \times w = 2^{23} \times 8 \quad \begin{cases} m = 23 \\ w = 8 \end{cases}$$

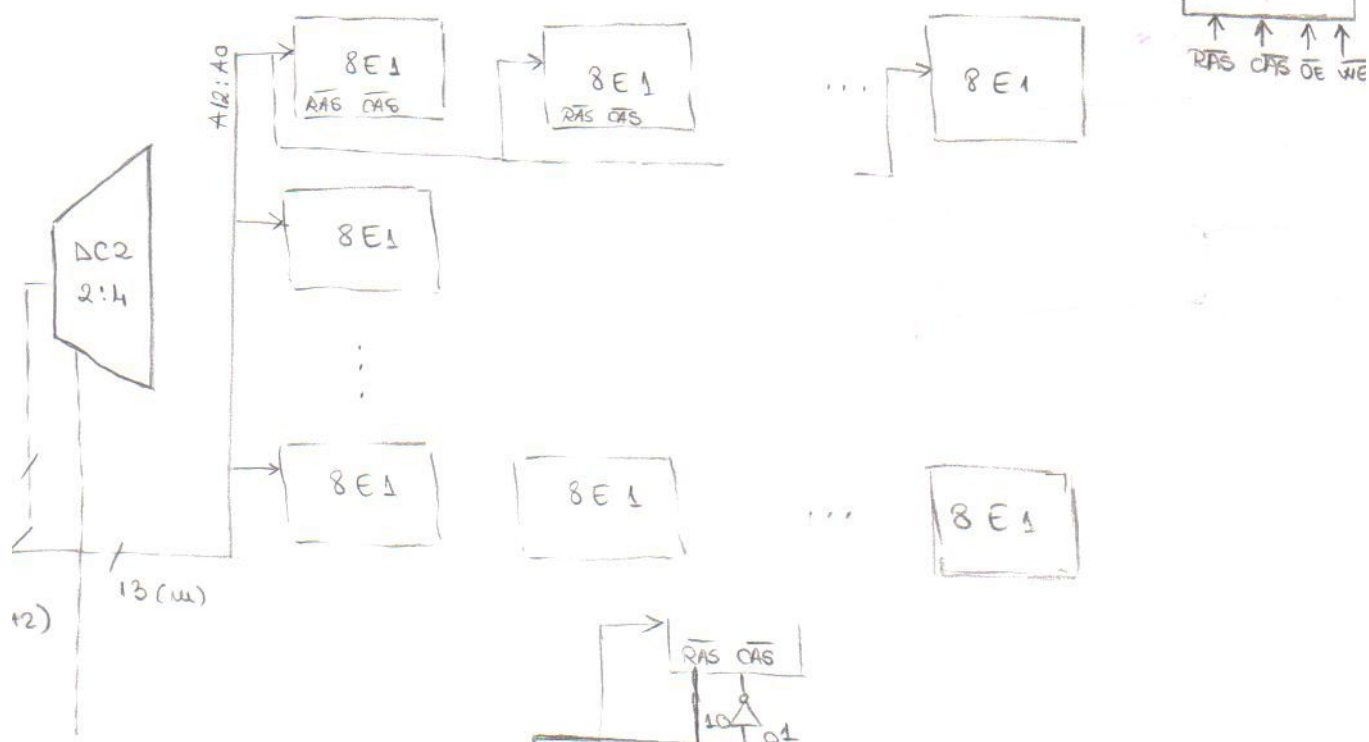
$$N = 2^{23} \quad N' = 2^{25} \quad p = \frac{2^{25}}{2^{23}} = 4$$

$$w = 8 \quad w' = 32 \quad q = \frac{32}{8} = 4$$

$p = 2^2 \rightarrow 2$ linii simplu.

$q = 4 \rightarrow$ cuv. creste de 4 ori

avem nevoie de $p \times q = 4 \times 4 = 16$ circuite 8E1



4.9.4.

Circ. DRAM 16 Mbiti are adim. cuv $w = 8$ biti. (ca $8 \in \mathbb{Z}$)
Organiz. 2D cu adr. multiplexată.

(a) Dacă adr. col = 10 biti, adr lin = ?

(b) câte circ. sunt neces. pt. o mem. $1G \times 32$ biti?

$$(a) 16 \text{ Mbiti} = 16 \times 2^{20} \text{ biti} = 2^4 \cdot 2^{20} = 2^{24}$$

$$2^m \times w = 2^{24} \Rightarrow 2^m = \frac{2^{24}}{w} = \frac{2^{24}}{8} = \frac{2^{24}}{2^3} = 2^{21} \text{ (linii adrese)}$$

m_x - rând

m_y - col.

$$m_x + m_y = 21 \Rightarrow m_x = 11 \text{ biti pt linie}$$

$$(b) 1G = 2^{10} \text{ M} = 2^{30}$$

$$\left. \begin{array}{l} N' = 2^{30} \quad N = 2^{21} \\ w' = 32 \quad w = 8 \end{array} \right\} \Rightarrow P = \left\lceil \frac{2^{30}}{2^{21}} \right\rceil = 2^9$$

$$2 = \frac{32}{8} = 4 = 2^2$$

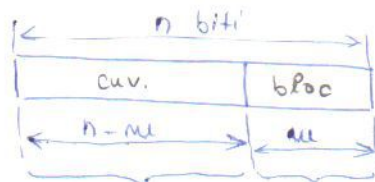
$$P \times 2 = 2^9 \cdot 2^2 = 2^{11} = 2048 \text{ circuite necesare}$$

4.9.6. Mem. DRAM $1M \times 16$ biti.

Utiliz. intercalarea cu 4 căi & 4 bancuri mem. M_0, M_1, M_2, M_3
Identif. bancul de mem. către care i se adesează frec. din.

(a) 01234h (b) ABCDEh (c) 91242h (d) FFFFFh

Adr. ucp



$$S_n = S_{n-m} + S_m$$

cuv. cu adr. i $\xleftrightarrow{\text{assign.}}$ unit. M_j , unde $j = i \bmod 2^m$

"intercalare cu 2^m cai"

biti pt. unitati
 $\downarrow$
 m

↑
adr. cuv

(a) intercalare cu 4 cai = 2^2 cai

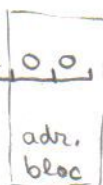
$$\text{hexa} = 16 = 2^4 \leftarrow \text{nr. biti}$$

$$A_{ucp} = 01234h$$

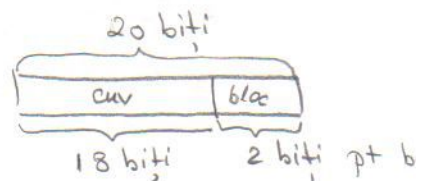
$$\rightarrow 5 \times 4 \text{ biti} = 20 \text{ biti}$$

adresa

0123 0100



$\Rightarrow$ blocul M_0



M_0	00
M_1	01
M_2	10
M_3	11

(b) $ABCDE h = ABCD \begin{array}{|c|c|c|c|} \hline 1 & 1 & 1 & 0 \\ \hline 3 & 2 & 1 & 0 \\ \hline \end{array}$
 $\Rightarrow$ bloc 10 = M_2

$E = 14 = 8 + 4 + 2 = 2^3 + 2^2 + 2^1$

(c) $91272 h = 9127 \begin{array}{|c|c|c|c|} \hline 0 & 0 & 1 & 0 \\ \hline \end{array}$ $\Rightarrow$ bloc 10 = M_2

(d) $FFFF h = FFFF \begin{array}{|c|c|c|c|} \hline 1 & 1 & 1 & 1 \\ \hline \end{array}$ $\Rightarrow$ bloc 11 = M_3

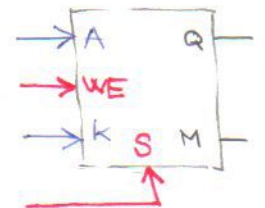
4.9.7. Utiliz. mem. asoc. din fig. 4.33(b) proiect. o unit. de mem. asoc. 4×4 biti.

$$M_i = \prod_{j=0}^{n-1} (A_j W_{ij} + \bar{A}_j \bar{W}_{ij} + \bar{K}_j)$$

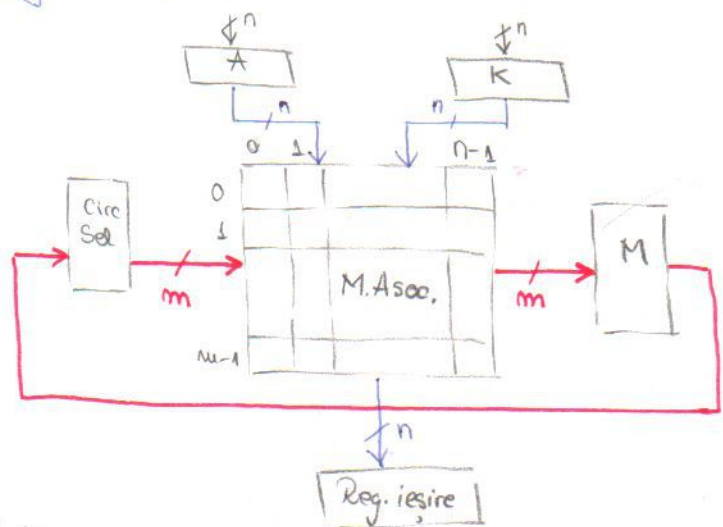
M_i = bitul masca

W_{ij} = bitul j al cuv. i din mem.

A_j, K_j = bitul j al neg. A & K



ex: m cuvinte $\times n$ biti
 A, R sunt de n biti
 M m biti



rezolv.

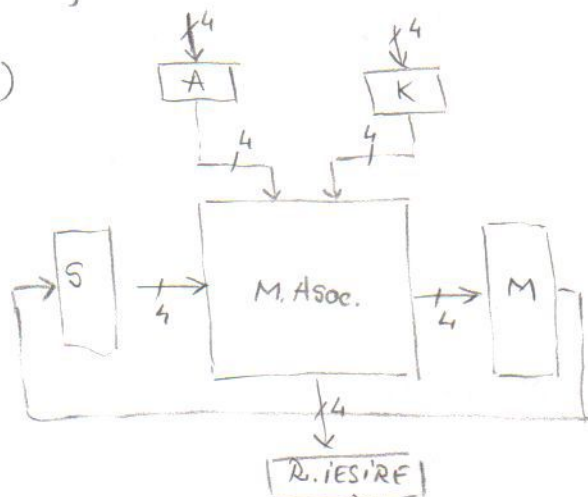
4×4 biti = 4 cuvinte $\times 4$ biti $\Rightarrow 16$ circuite
 A, R de 4 biti
 M de 4 biti

$$M_0 = \prod_{j=0}^{4-1} (A_j W_{0j} + \bar{A}_j \bar{W}_{0j} + \bar{K}_j)$$

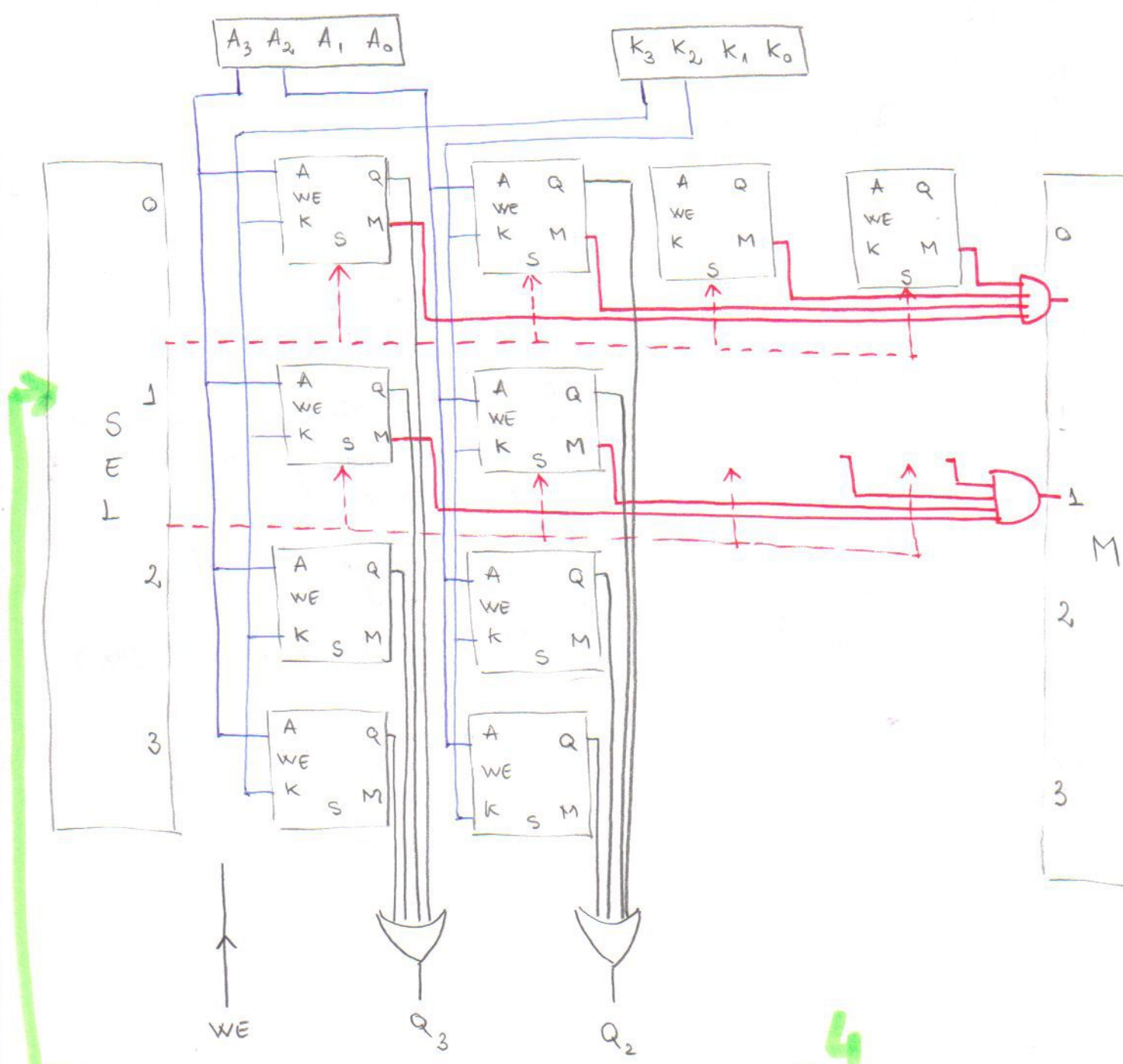
$$= (A_1 W_{01} + \bar{A}_1 \bar{W}_{01} + \bar{K}_1) \cdot$$

$$(A_2 W_{02} + \bar{A}_2 \bar{W}_{02} + \bar{K}_2) \cdot$$

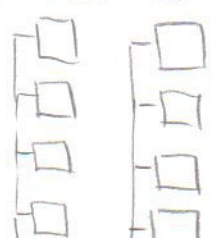
$$(A_3 W_{03} + \bar{A}_3 \bar{W}_{03} + \bar{K}_3)$$



PROBLEME

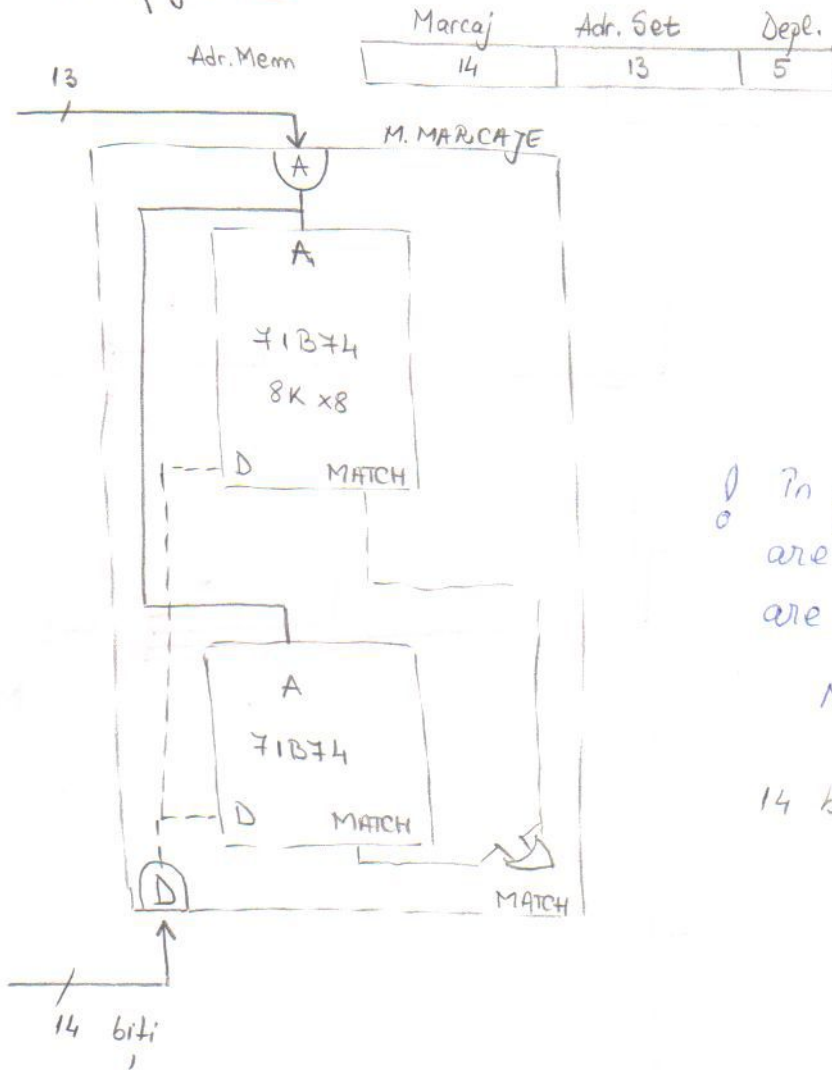


WE - conectat la toate simultan

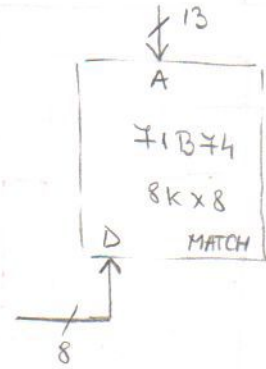


4.9.84

Schema la niv de registre pt. mem. RAM de marcare IDT 714 din fig 4.42



$$8K = 2^3 \cdot 2^{10} =$$



! In fig. 4.42 cur. marcat
are 14 biti pt. ca așa
are nevoie

M. Maroje 8k x 14

14 biti cuv $\Rightarrow$ 2 circ. 8.

4.15 M. cache cu map. directă (ex. 4.7.4.2. fig 4.42)
 Urmat. modif.: - capacitat. 64 kB
 - dim. bloc & Mag. Date 32 biti

- capacitat. 64 kB = $2^m \times w$

- dim. bloc = 32 biti = $2^5 \Rightarrow d = 5$

- dim. M.D. = 32 biti

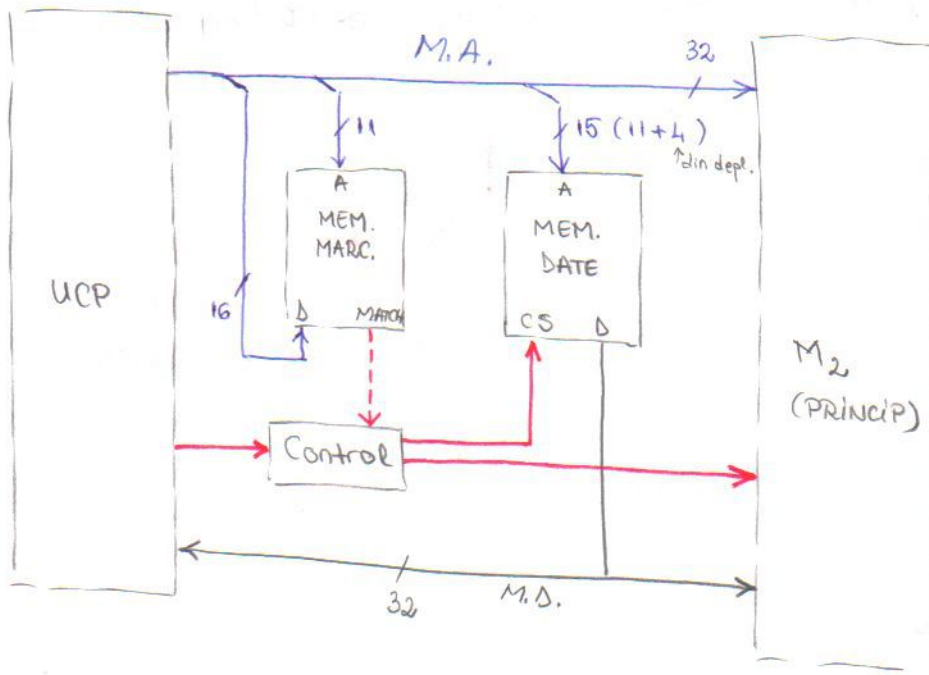
16	11	5
t	s	d

- nr. blocuri = $\frac{\text{capacitat.}}{\text{dim. bloc}} = \frac{2^{16}}{2^5} = 2^{11}$ blocuri

$s = 11$

- mem. marcare $2^{11} \times t$ biti

- M.A. = 32 biti $32 = t + s + d = t + 11 + 5 \Rightarrow t = 16$



M. DATE 4×71256

$\uparrow$

Mem. DATE $w' = 32 \text{ biti} = 2^5$

circ. $71256 : 32k \times \frac{8}{w}$

$\frac{w'}{w} = \frac{32}{8} = 4 \Rightarrow$ cuv. creste de 4 ori

$32k = 2^5 \cdot 2^{10} = 2^{15}$ Adr. cuv pe 15 biti

(folosim cei 4 ce lipsesc din depl.) -25-

! dim. bloc $\rightarrow$ deplasam
 $2^d \Rightarrow d$ deplasam

$\frac{\text{capacitat.}}{\text{dim. bloc}} = \text{nr. blocuri}$
 $2^5 \Rightarrow 5$

$t = \text{M.A.} - s - d$

9.16. Mem.
Cache

capacit : 2kB (mapare cu seturi asoc.)

16 seturi - fiecare 4 blocuri (linii)

Dim. adr. mem 32 biti, unit. de adresare 1 octet.

(a) Cărui set $\in$ 000010AFh

(b) Dacă 000010AFh & FFFFxyz h $\in$ același set.
ce val pot avea cifrele xyz

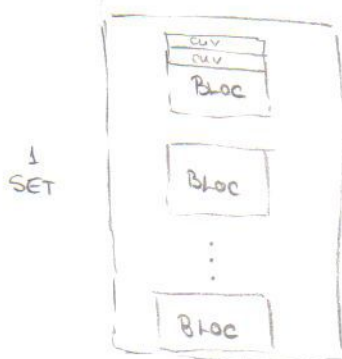
Map. cu seturi asoc.

Mem = $b = 2^5$ regiuni numite seturi $M_1(0), M_1(1) \dots M_1(b)$

Set = $k = 2^m$ blocuri

Mem. cache cu k blocuri mem. $\rightarrow$ k căi

(a) 16 seturi cu câte 4 blocuri



4 blocuri $\Rightarrow$ 4 căi

! Atentie

Mem = b seturi

Set = k blocuri

Bloc = cuvinte

Adr. mem = 32 biti

$$\frac{\text{capacitate}}{\text{căi} \times \text{dim. bloc}} = \frac{2 \cdot 2^{10}}{4 \times 2^5} = 2^4 = 16 \text{ seturi}$$

16 seturi $\times$ 4 blocuri = 64 blocuri

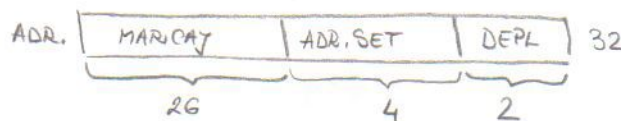
$$\text{dim. bloc} = \frac{2 \text{ k}}{64} = \frac{2 \cdot 2^{10}}{2^6} = 2^5$$

16 seturi = 2^4 seturi $\Rightarrow$ 4 linii pt. set

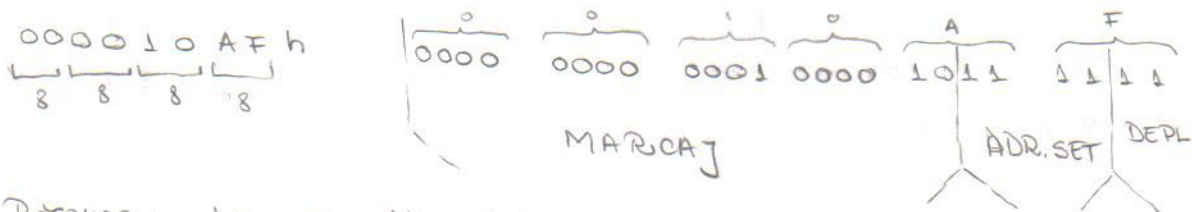
$$\text{dim. bloc} = 2^5 = 2^2 \cdot 2^3 = 2^2 \text{ octeti}$$

• pt. a identifica un octet. avem nevoie de un deplasare de 2

! $\frac{\text{capacit.}}{\text{căi} \times \text{dim. bloc}} = \text{nr. seturi}$
nr. blocuri într-un set



$$32 - 4 - 2 = 26$$



Răspuns: Adr. ∈ ultimului set (1111)

(b) FFFF7x|yz h ∈ setului 1111

FFFF7x | y₃y₂y₁y₀ z₃z₂z₁z₀

1 1 1 1 DEPL.

X = 0 : F (orice)

y = 00 11 3 h
01 11 7 h
10 11 A h
11 11 F h

z = 11 00 B h
11 01 C h
11 10 E h
11 11 F h

4.9.17

M. cache cu seturi asoc. cu 4 căi

Calc. cu dim. mem. fizice e 2^{32} octeți.

Dim. bloc 16 octeți și 2^{10} blocuri / set.

Calcul dim. mem. cache și lung. marajului?

• M. Fizică : - dim. = 2^{32} octeți = $2^{32} \times 8$

→ M.A. = 32 biți

• M. Cache : - dim. bloc = 16 octeți

- nr. blocuri = 2^{10} per set

→ $16 = 2^4 \Rightarrow$ 4 biți dep

dim. mem. cache = b = 2^5 seturi

set = k blocuri = 2^m blocuri = 2^{10} blocuri

→ {m = 10
LUNG. MA

M.A = m + s + d

32 = 10 + s + 4

s = 18

dim. mem. cache

b = $2^5 = 2^{18} = \dots$

4.9.18

Proiect. memorie cache cu seturi asoc. cu 4 căi (ex. sect. 4.4.4.3 p. 4.45)

unmă. param: capacitat. m. cache 64 KB
dim. bloc, 32 octeti (32 B)
M.D. 32 biti

• 4 căi $\Rightarrow k = 4$

• $\frac{\text{Capacitat.}}{\text{căi} \times \text{dim. bloc}} = b$

$$\frac{64 \text{ KB}}{4 \times 32 \text{ B}} = b$$

1 set

$$b = \frac{2^8 \cdot 2^{10}}{2^2 \cdot 2^5} = 2^9$$

$s = 9$

k : nr. blocuri

$k = 2^m$

b : nr. seturi

$b = 2^s$

m : marcaj

s : adr. set

d : deplasare

• dim. bloc 32 octeti $\rightarrow 2^5 \Rightarrow d = 5$

presup. că M.A. = 32 biti

$32 = m + s + d$

$m = 32 - 9 - 5 = 18$

$m = 18$

IMPLEMENTARE:

- In. ex. 4.45 pt. marcare s-au folosit: 2 RAM 512 x ^m20 biti
date 2 RAM 512 x 64 biti
MD

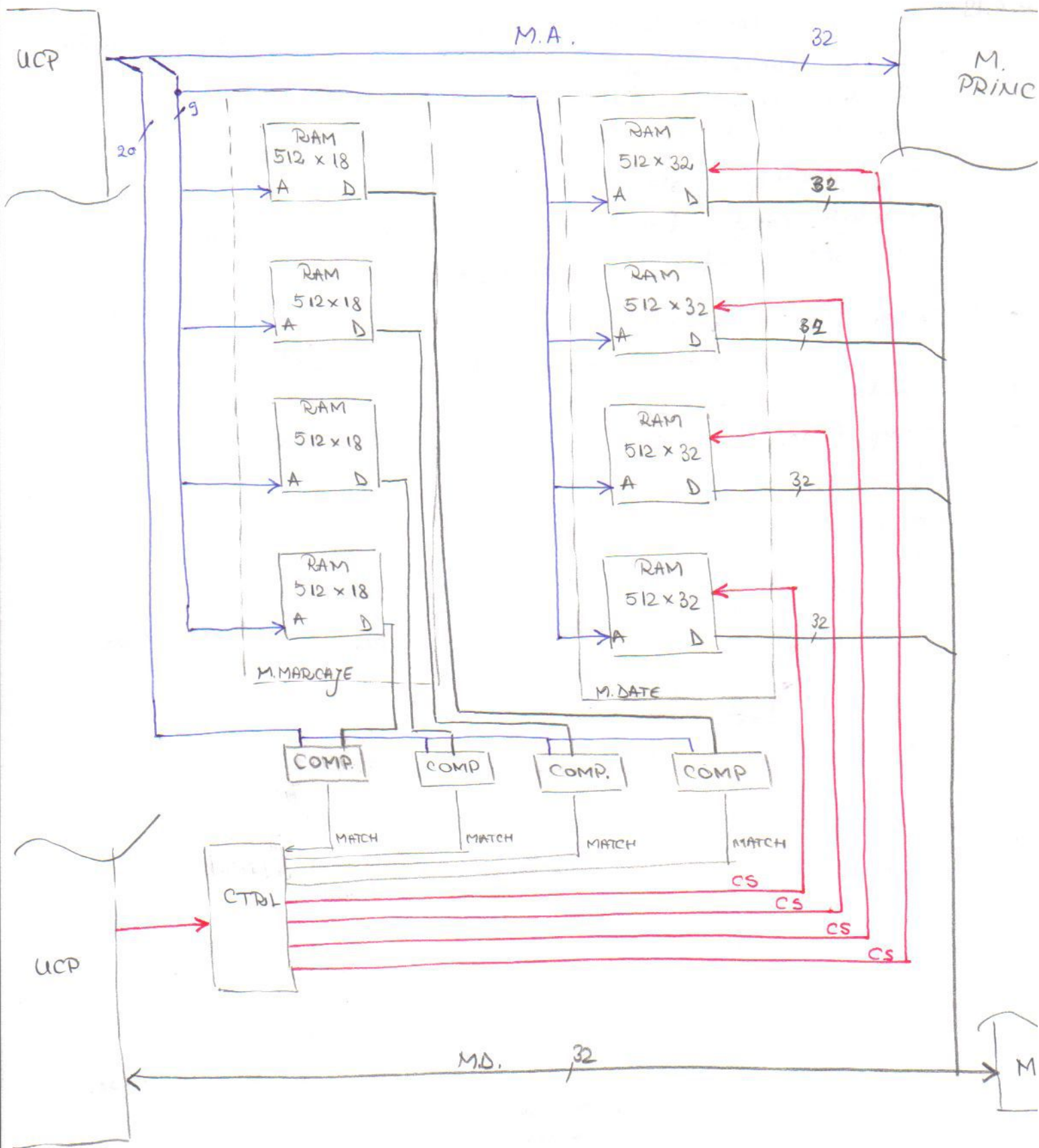
- noi avem nevoie de:

$512 = 2^9$

pt. MARCAJE : $2^5 \times m$ RAM
 $2^9 \times 18$
 512×18 biti

$\rightarrow 4 \times \text{RAM } 512 \times 18$

pt. DATE : $2^s \times \text{M.D.}$ RAM $\rightarrow 4 \times \text{RAM } 512 \times 32$
 $2^9 \times 32$ biti
 512×32



COMP: Comparator, marcaje
 CTRL: Logica control cit./scdere

4.4.19 Fie 3 m. cache de dim. reduse 4 blocuri cu câte un, cuvânt

1. m. cache complet asoc.
2. m. cache cu map. directă
3. m. cache cu seturi asoc. cu 2 căi

Pp. strateg. înloc. LRU.

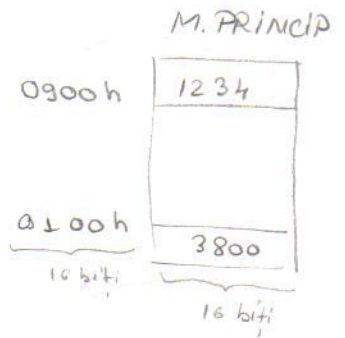
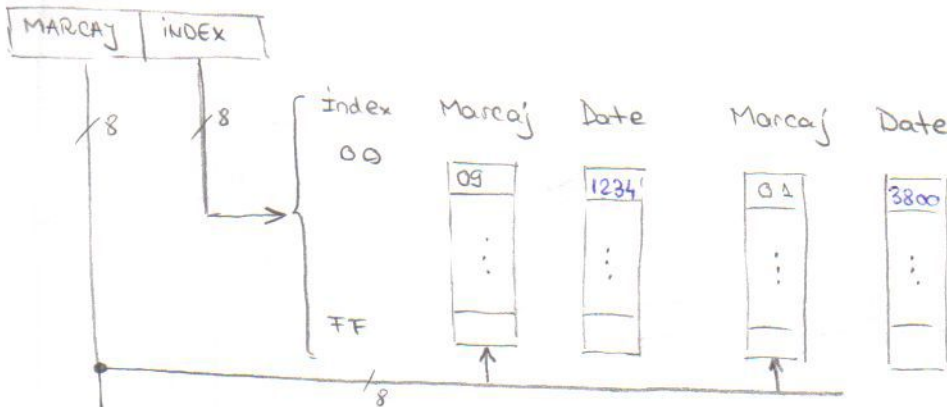
Arătați cont. fiec. mem. & determ. nr. eșecuri
pt. secu. refer. 0, 8, 0, 6, 8

- M. Seturi Asoc. $b = 2^9$ seturi
 $k = 2^m$ blocuri (căi)

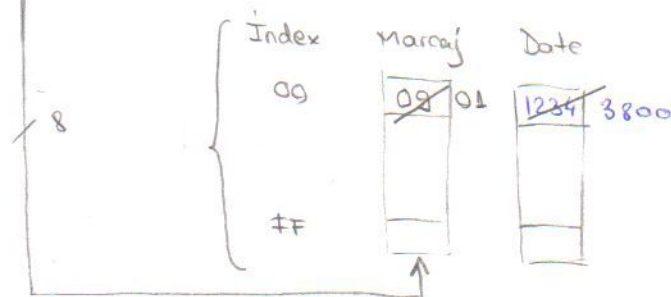
- M. Directă $k = 1$

- M. Comp. Asoc. $b = 1$

01 00 h
09 00 h

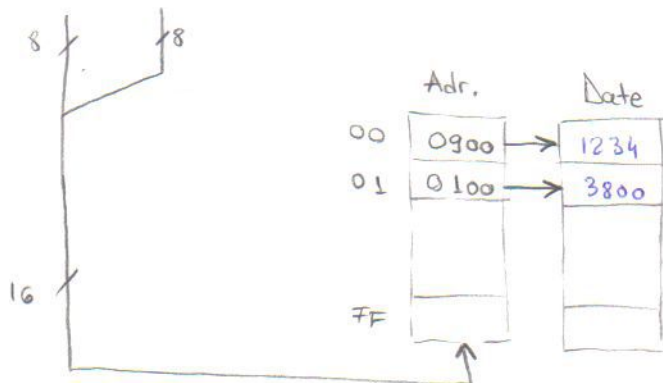


M. Seturi Asoc.
cu 2 căi



M. Directă

- marcajul și datele sunt înlocuite



M. Complet. Asoc.

PROBLEME

M. SETURI, ASOC.

Secv. 0 8 0 6 8
 Timep 1 2 3 4 5

0*	0*	0	6	6
	8	8*	0	0*
			8*	8
F	F	S	F	S

0 = 00h
 8 = 08h
 6 = 06h

LRU: blocul ce nu a
 fost utilizat de
 mai mult timp

$$F = 3$$

$$S = 2$$

$$PF = \frac{F}{F+S} = \frac{3}{5} (60\%)$$

④ Proiect. DRAM cu capacitat. $8M \times 8$ biti. Utiliz. circ. cu org. $2M \times 4$ biti. Desen. schema.

$$N' = 8M = 2^3 \cdot 2^{20} = 2^{23}$$

$$N = 2M = 2^1 \cdot 2^{20} = 2^{21}$$

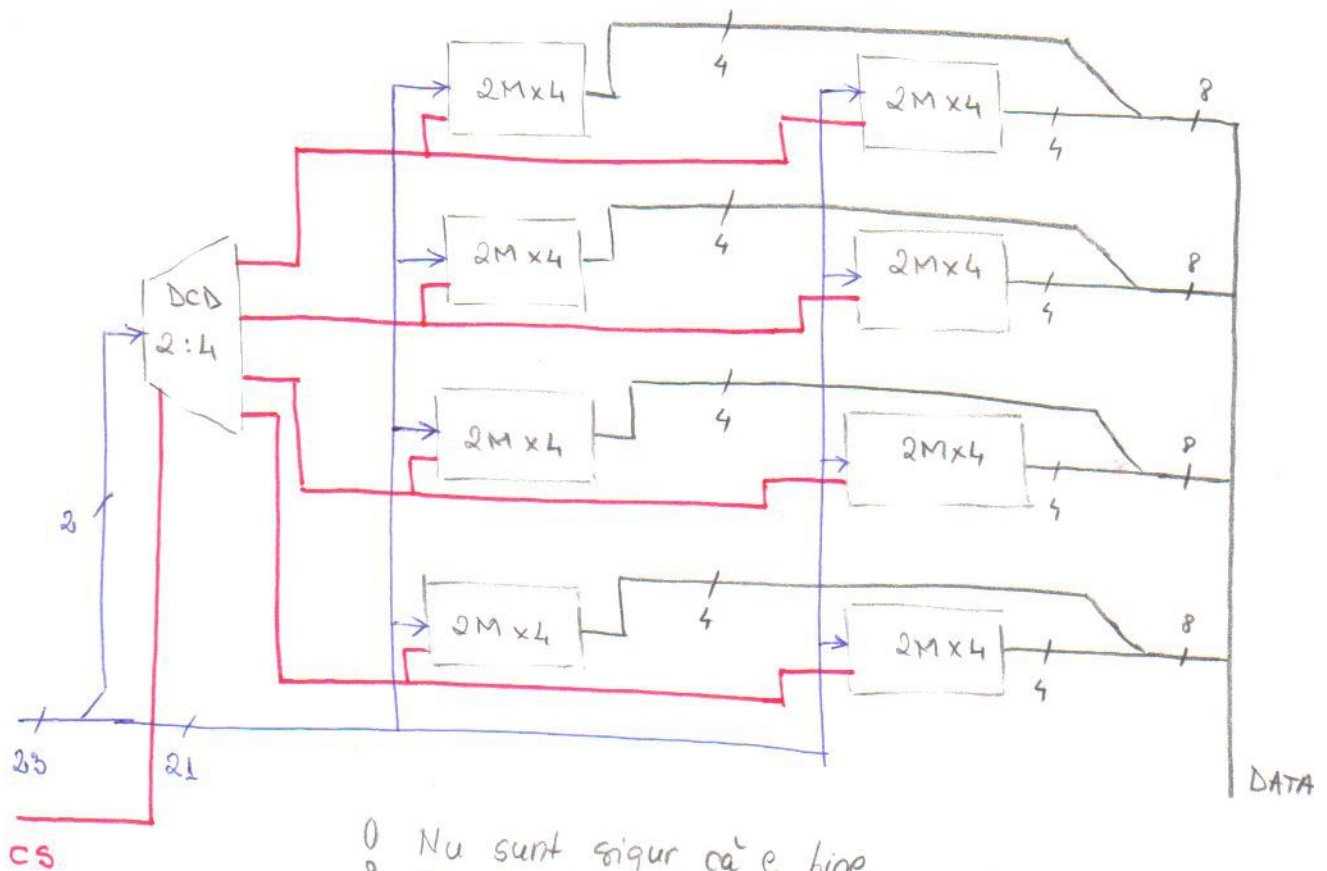
$$w' = 8 \text{ biti} = 2^3$$

$$w = 4 \text{ biti} = 2^2$$

$$p = \frac{2^{23}}{2^{21}} = 4 \quad (\text{2 linii supliment.})$$

$$q = \frac{2^3}{2^2} = 2 \quad (\text{cuv. crește de 2})$$

- avem nevoie de $p \times q = 4 \times 2 = 8$ circuite de memorie $2M \times 4$ biti



! Nu sunt sigur că e bine.
Se cere memor. DRAM și nu RAM.
Ar mai fi nevoie de RAS / CAS

④ Utiliz. ca bază o memor. $8M \times 8$ biți.

Proiect. DRAM $32M \times 32$ biți. Desen.

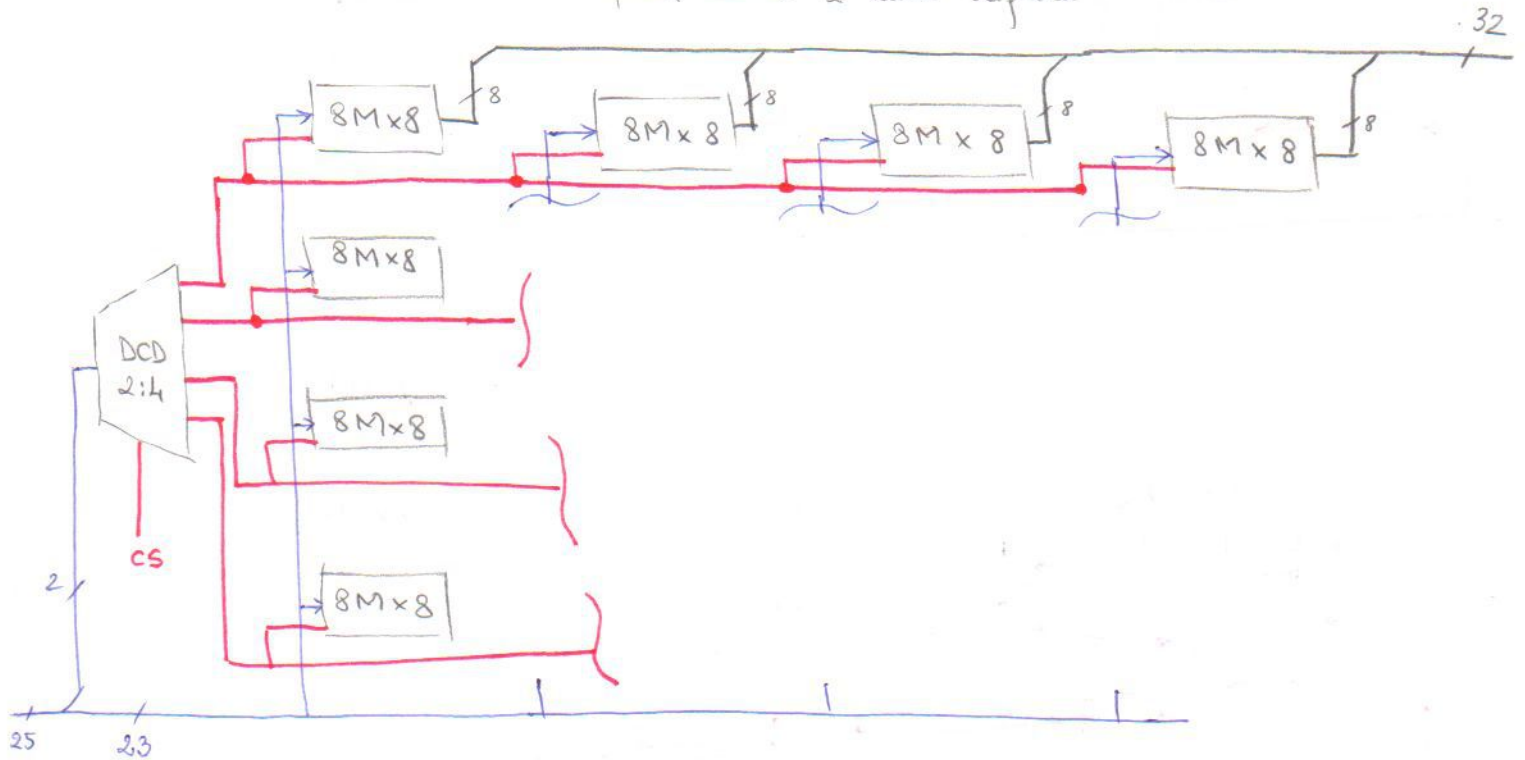
$$p = \frac{32M}{8M} = 4$$

$$\bullet \text{ nr. circuite } p \times q = 16$$

$$q = \frac{32}{8} = 4$$

$$8M = 2^3 \cdot 2^{20} = 2^{23} \longrightarrow 23 \text{ linii adresă}$$

$$p=4 \longrightarrow 2 \text{ linii suplimentare}$$



⑤ Proiect m. cache cu map. directă: capac. $8KB$, dim. bloc 8 octet, $MA = 32$ biți, M.D. 64 biți, Struct. cuv. adr. ? Desen ?

$$\bullet \text{ aflăm nr. blocuri } \frac{8KB}{8B} = 1K = 2^{10} \longrightarrow \boxed{s=10}$$

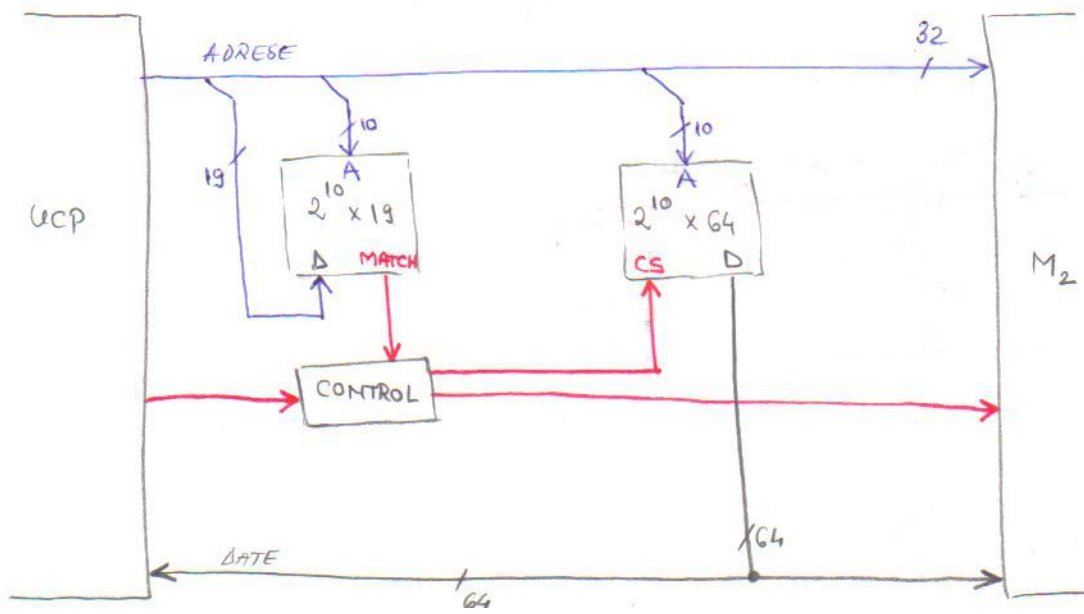
$$\bullet \text{ aflăm deplasare } 8B = 2^3 B \longrightarrow \boxed{d=3}$$

$$MA = s + t + d \longrightarrow \boxed{t=19}$$

$$32 = 10 + t + 3$$

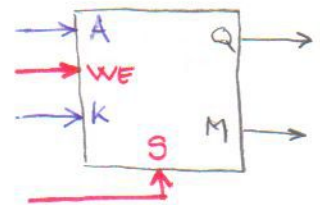
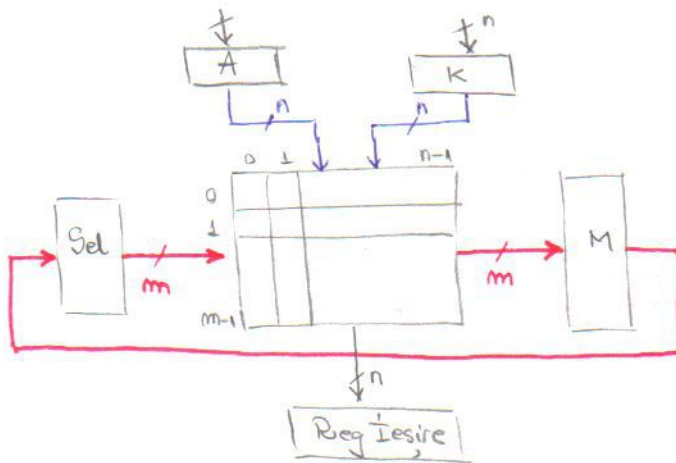
Adr.		
Marcaj	Index	Depl
19	10	3

• circuite:
 $\rightarrow \text{MARCAJ: } 2^s \times t$
 $\rightarrow \text{DATE: } 2^s \times 64$

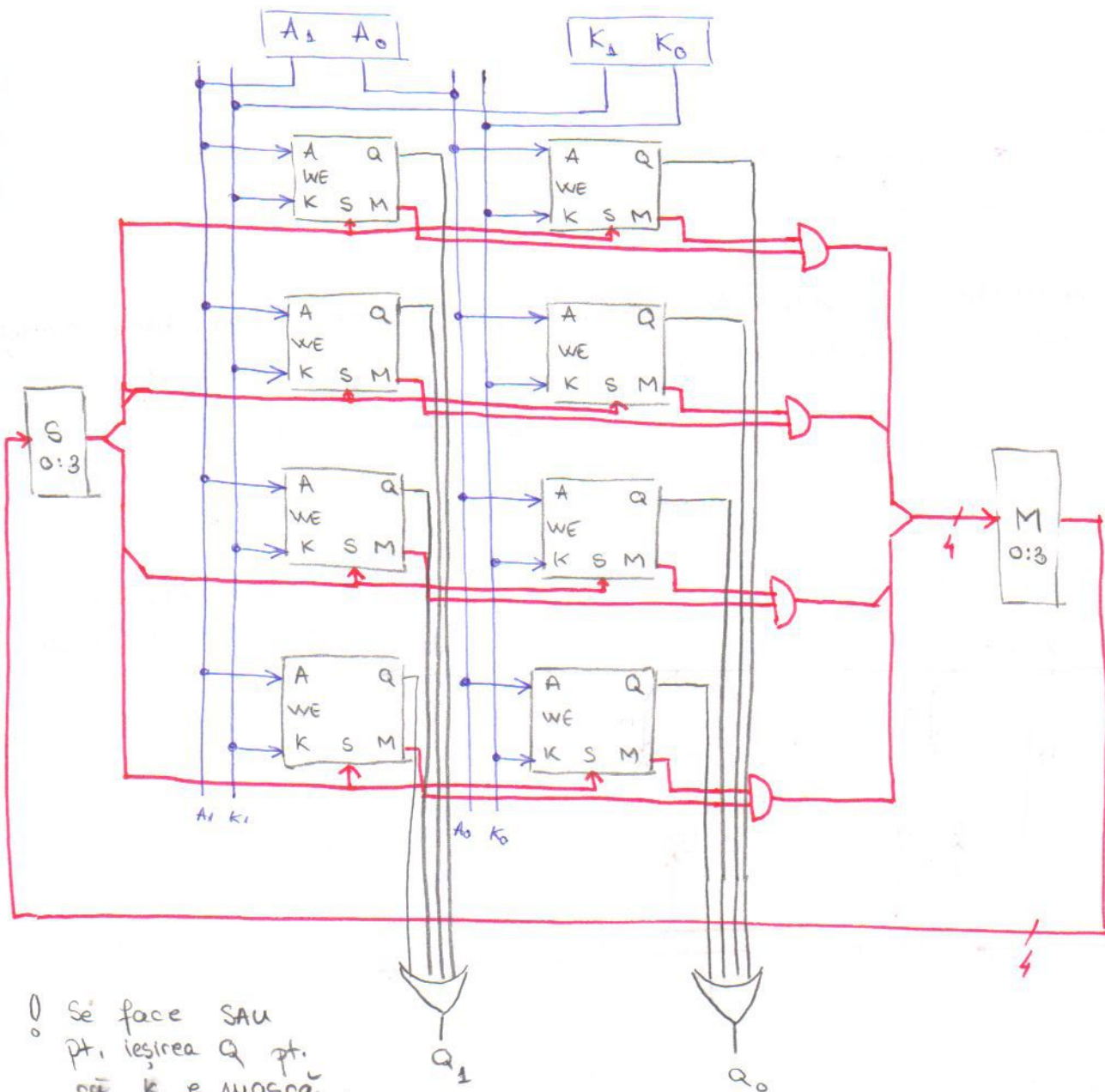


7) using the cell from the lecture. Design an assoc. mem. with 4 words of 2 bits each.

m cuvinte $\times n$ biti



celula din curs



! Se face SAU
pt. iesirea Q pt.
ca K e masca.

PROBLEME

FIFO

ex:

Time
Secv

1 2 3 4 5 6 7 8 9 10 11
0 1 2 0 3 2 0 1 2 4 0

n=3 cadre

0*	0*	0*	0*	3	3	3	3*	2	2	2*
	1	1	1	1*	1*	0	0	0*	4	4
		2	2	2	2	2*	1	1	1*	0
F	F	F	S	F	S	F	F	F	F	F

F=9 S=2 PF= 9/11 ~ 0,81

n=4 cadre

0	1	2	0	3	2	0	1	2	4	0
0*	0*	0*	0*	0*	0*	0*	0*	0*	4	4
	1	1	1	1	1	1	1	1	1*	0
		2	2	2	2	2	2	2	2	2
				3	3	3	3	3	3	3
F	F	F	S	F	S	S	S	S	F	F

F=6 S=5 PF= 6/11 ~ 0,54

"Cel mai putin recent incarcat
(ultimul incarcat)

FIFO →

0				
1	0			
2	1	0		
3	2	1	0	
4	3	2	1	0
0	4	3	2	1

LRU

ex:

Time
Secv

1 2 3 4 5 6 7 8 9 10 11
0 1 2 0 3 2 0 1 2 4 0

0*	0*	0*	0	0	0*	0	0	0*	4	4
	1	1	1*	3	3	3*	1	1	1*	0
		2	2	2*	2	2	2*	2	2	2
F	F	F	S	F	S	S	F	S	F	F

Secv.

0	1	2	0	3	2	0	1	2	4	0
0*	0*	0*	0	0	0	0	0	0	0*	0
	1	1	1*	1*	1*	1*	1	1	1	1*
		2	2	2	2	2	2*	2	2	2
				3	3	3	3	3*	4	4
F	F	F	S	F	S	S	S	S	F	S

"Care nu a fost utilizat
de cel mai mult times

F=4 S=4
PF= $\frac{4}{11} \sim 0,36$

F=5 S=6
PF= $\frac{5}{11} \sim 0,45$

10) M. virt. - 4 cadre

- secv 1, 4, 10, 4, 3, 6, 1, 1, 8, 6, 9, 4, 3, 3, 3, 1

- FIFO

- PF = ?

! Observatie FIFO cand apare FAIL atunci se modif. blocul selectat.

Secv.

1	4	10	4	3	6	1	1	8	6	9	4	3	3	3	1
1*	1*	1*	1*	1*	6	6	6	6	6	6*	4	4	4	4	4
	4	4	4	4	4*	1	1	1	1	1	1*	3	3	3	3
		10	10	10	10	10*	10*	8	8	8	8	8*	8*	8*	1
				3	3	3	3	3*	3*	9	9	9	9	9	9*
F	F	F	S	F	F	F	S	F	S	F	F	F	S	S	F

$$F = 11 \quad S = 5 \quad PF = \frac{11}{16}$$

F = fail = bloc incarc. in memor. virt.

11) M. virt. - 4 cadre

- secv. aceeași → 10

- LRU

- PF = ?

! Observatie LRU cand este Success atunci e selectat alt bloc.

(PARTIAL
CORRECT)

S = succes = acces la bloc. in memor.

FIFO

1	4	10	4	1	3	10	4	6	3	10	1	6	3	10	8	1	6	3	9	8	1	6	4	9	8	1	3	4	8	8	1	3	4	9
---	---	----	---	---	---	----	---	---	---	----	---	---	---	----	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Secv.

1	4	10	4	3	6	1	1	8	6	9	4	3	3	3	1
1*	1*	1*	1*	1*	6	6	6	6	6	6	6	6*	6*	6*	1
	4	4	4	4	4*	1	1	1	1	1	1*	4	4	4	4
		10	10	10	10	10*	10*	8	8	8	8*	3	3	3	3
				3	3	3	3	3*	3*	9	9	9	9	9	9*
F	F	F	S	F	F	F	S	F	S	F	F	F	S	S	F

$$F = 11 \quad S = 5 \quad PF = \frac{11}{16} = \frac{1}{4} \cdot \frac{11}{4} = \frac{2,75}{4} \approx 0,7$$